

**OIL PALM FRESH FRUIT BUNCH MASS ESTIMATION USING RGB-
DEPTH DATA**

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**BACHELOR OF AGRICULTURAL AND BIOSYSTEMS ENGINEERING
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DEPTH DATA**

By

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**Report Submitted to the Faculty of Engineering, Universiti Putra Malaysia,
in Partial Fulfillment of the Requirement for the
Bachelor of Agricultural and Biosystems Engineering with Honours**

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PROJECT APPROVAL FORM

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ABSTRACT

Oil palm is a major agricultural crop in Malaysia, and accurate estimation of Fresh Fruit Bunch (FFB) mass is important for improving yield prediction and plantation management. However, current methods rely on manual weighing, which is time-consuming and labour-intensive. This study investigates the feasibility of using RGB-Depth (RGBD) data as a non-contact method to estimate the volume of oil palm FFB, which can then be used to estimate mass based on density. In current plantation practice, FFB mass is used to plan harvest and oil extraction, but measuring mass typically requires lifting the fruit onto a weighing scale. Since no non-contact method exists, this study explores a volume-based approach using RGBD sensing. A total of 50 FFB samples were collected at Havys Oil Mill Sdn. Bhd., and RGBD videos were recorded using an Intel RealSense D435i camera mounted in a top-down position. RGBD combines RGB (color) images with depth data to provide 3D spatial information. Physical mass and volume were measured using a mechanical spring scale and the water displacement method. Point cloud data was extracted from the depth video and processed in CloudCompare using segmentation, normal computation, and 2.5D volume estimation. Average density was calculated from the measured mass and volume, then applied to convert estimated volume into mass. Statistical analysis showed a strong correlation between estimated and actual mass. The results suggest that RGBD imaging can support real-time, non-contact FFB mass estimation, and could be integrated into oil palm grabbers to improve harvest planning and reduce processing delays.

ABSTRAK

Kelapa sawit merupakan tanaman pertanian utama di Malaysia, dan anggaran jisim Tandan Buah Segar (TBS) yang tepat penting bagi ramalan hasil dan pengurusan ladang. Namun, kaedah semasa bergantung kepada penimbangan manual yang memakan masa dan memerlukan tenaga kerja. Kajian ini menilai kebolegunaan data RGB-Depth (RGBD) sebagai kaedah tanpa sentuhan untuk menganggarkan isipadu TBS, yang kemudiannya digunakan untuk mengira jisim berdasarkan ketumpatan. Dalam amalan semasa, jisim TBS digunakan untuk merancang penuaian dan pengekstrakan minyak, tetapi pengukuran jisim memerlukan sentuhan fizikal. Oleh itu, pendekatan anggaran jisim melalui isipadu dicadangkan. Sebanyak 50 sampel TBS telah dikumpul dari Havys Oil Mill Sdn. Bhd., dan video RGBD dirakam menggunakan kamera Intel RealSense D435i yang dipasang dari sudut atas. RGBD menggabungkan imej berwarna (RGB) dan data kedalaman (depth) untuk menghasilkan maklumat bentuk tiga dimensi. Jisim dan isipadu sebenar diukur menggunakan penimbang spring mekanikal dan kaedah sesaran air. Data point cloud diekstrak daripada video kedalaman dan diproses dalam perisian CloudCompare melalui pemisahan objek, pengiraan normal, dan anggaran isipadu 2.5D. Ketumpatan purata dikira daripada jisim dan isipadu sebenar, dan digunakan untuk menukar isipadu kepada jisim anggaran. Analisis statistik menunjukkan korelasi yang kuat antara jisim anggaran dan sebenar. Hasil kajian menunjukkan bahawa RGBD boleh digunakan sebagai kaedah tanpa sentuhan untuk anggaran jisim TBS secara masa nyata, dan berpotensi untuk diintegrasikan pada jentera pengambil TBS bagi meningkatkan perancangan penuaian dan mengurangkan kelewatan pemprosesan.

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LIST OF ABBREVIATIONS

2D	Two-Dimensional
3D	Three-Dimensional
AI	Artificial Intelligence
ANN	Artificial Neural Network
AR	Augmented Reality
CNN	Convolutional Neural Networks
FFB	Fresh Fruit Bunch
FOV	Field of View
fps	Frames per Second
IoT	Internet of Things
IQR	Interquartile Range
IR	Infrared
LiDAR	Light Detection and Ranging
ML	Machine Learning
MSPO	Malaysian Sustainable Palm Oil
PCL	Point Cloud Library
r	Pearson Correlation Coefficient
r ²	Coefficient of Determination
RGB-D	Red Green Blue - Depth
RM	Ringgit Malaysia

RMSE	Root Mean Square Error
ROS	Robot Operating System
SfM	Structure-from-Motion
ToF	Time-of-Flight
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
VR	Virtual Reality
WDM	Water Displacement Method

CHAPTER 1

INTRODUCTION

1.1 Background of Studies

Oil palm (*Elaeis guineensis*) is a major agricultural crop, especially in Southeast Asia, Africa, and South America, where its fresh fruit bunches (FFB) are processed into palm oil a commodity widely used in food, cosmetics, and biofuel industries (Hashemvand Khiabani & Takeuchi, 2020). Countries such as Indonesia and Malaysia dominate global palm oil production, making the crop vital to their economies.

Traditionally, yield estimation in oil palm plantations depends on manual methods, where workers visually inspect, count, and sometimes weigh FFB to estimate average bunch mass (Corley & Tinker, 2015). These conventional practices are labor-intensive, time-consuming, and prone to human error, limiting their scalability and consistency, especially in large plantations (Shiddiq et al., 2022). Accurate estimation of FFB mass is crucial because it directly influences the calculation of palm oil yield, which is a higher FFB mass correlates with a greater amount of extractable oil, though oil content can vary with fruit quality and ripeness (Corley & Tinker, 2015).

Recent advancements in imaging technologies, particularly RGB-Depth (RGBD) sensors, offer a promising alternative for automated yield estimation. RGBD imaging combines traditional color (red, green, blue) data with depth information, producing a three-dimensional representation of objects, which allows for more accurate measurement of size, shape, and spatial features compared to standard color images (Chua et al., 2021). In agriculture, RGBD imaging has been successfully used

for fruit counting, plant phenotyping, and biomass estimation in crops such as apples and grapes (Lomte et al., 2019)

The key advantage of RGBD data in mass estimation lies in its ability to capture depth information, enabling the construction of a 3D model of the FFB. From this model, the volume of the bunch can be estimated using computational algorithms, and with known or measured density values, the mass can be predicted using the relationship, $\text{Mass} = \text{Volume} \times \text{Density}$ (Pamornnak et al., 2017). This approach reduces manual intervention, minimizes errors, and can be scaled across large plantations, especially when combined with machine learning techniques such as convolutional neural networks (CNNs) for automated feature extraction and improved prediction accuracy (Chua et al., 2021).

However, challenges remain, as oil palm FFB are large, irregularly shaped, and often partially hidden by leaves, making accurate volume and mass estimation difficult under varying environmental conditions like lighting and occlusion (Zhang et al., 2020). Although preliminary studies have shown the feasibility of using RGBD imaging for FFB volume and mass estimation, many lack robust field validation or do not fully address real-world variability (Septiarini et al., 2019).

The success of estimating the mass of Fresh Fruit Bunches (FFB) depends greatly on how accurately the volume and size of the fruit can be measured. A well-designed volume measurement system is essential to capture the physical characteristics of FFB in a consistent and practical manner. By using RGB-Depth imaging technology, this study focuses on developing a system that can measure the

volume of FFB efficiently. In addition, this research aims to identify the correlation between the RGB-D data and the actual volume, mass, and size of the fruit. Understanding this relationship is key to building a foundation for future mass estimation models. The outcomes of this project are expected to improve data collection methods and support better yield estimation practices in oil palm plantations.

1.2 Problem Statements

In oil palm plantations, estimating the mass of Fresh Fruit Bunches (FFB) is important for managing yield and planning logistics. However, most current methods rely on manual weighing at collection points, which takes time and can lead to delays. These delays affect how quickly the fruits are processed, which may reduce their quality and overall oil extraction. To solve this, a contactless method and more automated method is needed. One possible solution is to estimate the volume of FFB using RGB-Depth (RGB-D) data, which captures both color and depth information of the fruit where the depth data is used to generate a 3D point cloud of the fruit, allowing volume estimation through surface reconstruction techniques. Estimating volume from RGB-D data could be a practical way to predict the mass without needing to weigh each bunch manually. However, it is still unclear how well RGB-D volume data relates to actual mass and volume measurements of FFB. To find out, it is important to determine the density of FFB and compare the estimated volume from RGB-D data with the actual measured volume. This study aims to investigate the feasibility of using RGB-D data to estimate the volume of oil palm FFB for the purpose of contactless method mass estimation. If

successful, this approach could support better decision-making in harvest planning, transportation, and processing. By enabling early mass estimation before harvesting, allowing plantation managers to assign the right number of workers and transport vehicles. During transportation, it helps in planning truck loads to avoid overloading or underloading. In processing, knowing the expected mass earlier can help the mill schedule operations and reduce waiting times, which helps maintain fruit freshness.

1.3 Objective

This research aims to investigate feasibility RGB-Depth data to estimate volume of Fresh Fruit Bunch (FBB) for contactless method of FFB mass estimation purposes.

The specific objectives are as follows:

- i. To determine the density of oil palm Fresh Fruit Bunch (FBB).
- ii. To determine the variations between volume estimated using RGBD sensing data and physically measured volume of oil palm FFB.

1.4 Scope of Study

The scopes of the project are:

- i. The study investigates the feasibility of using RGB-Depth (RGB-D) data to estimate the volume of oil palm Fresh Fruit Bunches (FFB).

- ii. The study aims to determine the density of oil palm Fresh Fruit Bunch (FFB) and To determine the variations between volume estimated using RGBD sensing data and physically measured volume of oil palm FFB.
- iii. A total of 50 FFB samples were used for this study, and physical measurements such as volume (via water displacement), mass (via mechanical scale), and size (via measuring tape) were collected as actual data.
- iv. Point cloud data was extracted from depth videos in Intel Realsense Viewer software and it processed the volume estimation using CloudCompare software.
- v. The research is limited to the use of a top-down RGB-D camera setup using the Intel RealSense D435i sensor in an outdoor environment at a palm oil mill loading ramp.
- vi. The study does not develop a complete mass prediction model using machine learning.
- vii. The research does not test performance under varied environmental conditions such as different lighting and occlusion. It only tests in bright and good weather only.
- viii. The large size and weight of FFBs present challenges in the data collection process.

1.5 Thesis Outline

This thesis consists of five chapters. Chapter 1 introduces the background of the study, the problem being addressed, and the motivation for exploring RGB-Depth (RGB-D) data as a tool for estimating the volume of oil palm Fresh Fruit Bunches (FFB). It also outlines the research objectives, scope of study, and significance of this work in the context of improving yield estimation and plantation management. Chapter 2 presents a literature review covering several key areas: the oil palm industry in Malaysia, the basics and applications of RGB-D and point cloud data, and previous studies on non-contact methods of weight estimation in agriculture. This chapter helps establish the relevance of RGB-D data and volume estimation techniques in agricultural practices, especially in oil palm operations. Chapter 3 describes the methodology used in this research. It includes the setup for RGB-D data collection using the Intel RealSense camera, the process for physical measurements of FFB volume and mass, point cloud extraction and segmentation, volume estimation using CloudCompare software, and the evaluation of results through statistical analysis. Chapter 4 will present and discuss the findings of the study. It will compare the estimated volume and mass from RGB-D data with actual measured values and evaluate the correlation. Finally, Chapter 5 will conclude the thesis by summarizing the key findings, contributions, and limitations of the study. It will also suggest possible improvements and future work to further develop automated volume and mass estimation systems for oil palm FFB using RGB-D data.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

This chapter presents a comprehensive review of the concepts, technologies, and methods relevant to the estimation of oil palm Fresh Fruit Bunch (FFB) mass using RGB-Depth (RGB-D) data. It begins by outlining the significance of the oil palm industry in Malaysia and the challenges faced in maintaining productivity and efficiency. The chapter then introduces RGB-D data, explaining how this technology captures both color and depth information to create detailed 3D representations, which are increasingly used in agriculture for tasks such as fruit volume measurement and plant phenotyping. The discussion extends to point cloud data, which provides spatial information in 3D form and supports advanced agricultural applications including canopy analysis, digital mapping, and automation. The integration of point cloud and RGB-D data is especially useful in estimating fruit volume and structure in oil palm plantations. Lastly, the chapter explores various non-contact weight estimation methods that rely on image processing, machine learning, and 3D data, demonstrating their effectiveness and efficiency in replacing manual weighing. These insights form the foundation for this project's approach in developing a system for estimating the mass of FFB using RGB-D and point cloud data.

2.2 Oil Palm Industries in Malaysia

The oil palm (*Elaeis guineensis*) is a tropical crop that originated in West Africa and has become a major part of Southeast Asia's agricultural economy. Introduced to

Malaysia in the early 20th century, it shifted from being an ornamental plant to a key commercial crop. Malaysia is currently the second-largest global producer of palm oil, contributing approximately 33% of the global supply (Kristanti et al., 2021). The industry has modernized over time, with advances in genetics, mechanization, and sustainability. Palm oil is widely used around 80% goes into food products like cooking oil, margarine, and snacks, while the rest supports non-food industries such as biodiesel and cosmetics. According to Parveez et al. (2021), the oil palm industry is a cornerstone of Malaysia's economy, contributing to GDP and export earnings. In 2021, it made up over 5% of GDP and supported rural development and smallholder entrepreneurship (Arifin et al., 2023).

Currently, FFB yield is estimated manually in most plantations, either by counting fruit bunches per tree or weighing them during collection. However, this method is time-consuming and often lacks accuracy, especially when done across large plantations. The weight of FFB is directly linked to crude palm oil (CPO) yield heavier, ripe bunches typically produce more oil. Therefore, fast and accurate mass estimation during harvest is critical for planning transport, processing schedules, and maximizing oil extraction efficiency.

Despite its economic strength, the industry faces several challenges. Labor shortages are a major issue, worsened by dependence on foreign workers and restrictions during COVID-19 (Naidu & Moorthy, 2021). Environmental concerns, especially deforestation, have also led to import restrictions in markets like the EU. To improve sustainability, Malaysia has introduced the Malaysian Sustainable Palm Oil (MSPO) certification to promote responsible practices and address international

criticism. According to Ahmad et al. (2020), oil palm is the world's most efficient oil crop, producing more oil per hectare than soy or sunflower. It provides income and jobs to millions, particularly in rural areas, and generated over RM 108 billion in export revenue in 2021. While manual labor remains common, it's less efficient and poses safety risks. Mechanization such as drones and automated harvesters is gradually improving operations but requires high investment (Ong-Abdullah et al., 2017). Crowley (2020) says that training local workers for technical roles can help address labor shortages while maintaining productivity.

In conclusion, the oil palm industry remains vital to Malaysia's economy. However, improving efficiency through better FFB yield estimation is essential. By adopting faster, automated methods and linking FFB mass to oil yield, the industry can better plan harvests, reduce waste, and support sustainable growth.

2.3 RGB-Depth Data

2.3.1 Introduction of RGB-Depth Data

RGB-Depth (RGB-D) refers to the fusion of traditional Red-Green-Blue (RGB) color images with depth data, offering a more comprehensive understanding of both visual and spatial aspects of a scene. Applications that demand precise geometry in addition to visual detail are made possible by this combination, which fills the gap between 2D imaging and 3D spatial representation. RGB-D technology has improved object detection, scene understanding, and 3D reconstruction jobs, which has had a substantial impact on a number of fields, including robots, virtual reality, and agriculture. Research on RGB-D applications in computer vision, robotics, and

healthcare exploded as a result. Later gadgets like Orbbec Astra and Intel RealSense, which provided better resolution and depth sensing methods, significantly developed the field (Henry et al., 2012).

There are 2 primary components of RGB-D systems. Firstly, RGB camera which captures traditional color images that provide visual details like texture and color gradients. The second is, Depth sensor. Depth sensor functionally measures the distance from the sensor to objects in the environment. Techniques include Time-of-Flight (ToF), which calculates the time it takes for a light pulse to return to the sensor, and structured light, which uses projected patterns to estimate depth.

Next, RGB-D sensors can be categorized based on their depth-sensing mechanisms which have two. Active sensors is one of the depth-sensing mechanisms that use infrared (IR) light or structured light patterns to calculate depth, as seen in Microsoft Kinect and Intel RealSense. Other than active sensors there are also passive sensors which rely on stereoscopic vision, comparing two slightly offset images to derive depth.

One of the key advantages of RGB-D systems is their ability to generate multi-dimensional datasets. These datasets provide more detail than conventional RGB or depth-only systems by combining depth measurements with pixel-level RGB colour values. In addition, Zhang & Funkhouser. (2018) also studied that this characteristic is crucial for tasks like 3D mapping, where precise measurements of object shapes and positions are required. Additionally, RGB-D systems have high-resolution sensing capabilities, which allow them to pick up on even the smallest spatial alterations.

Applications like plant phenotyping, where small variations in leaf development or structure must be tracked, benefit greatly from this characteristic. However, despite their advantages, RGB-D systems do have limitations. Reflective, clear, or far-off surfaces can be difficult for depth sensors, especially those that use infrared (IR) light, to detect, which can result in inaccurate data.

2.3.2 Process of RGB-Depth Data

Furthermore, Henry et al. (2012) studied the RGB colour imagery and depth information are the two main forms of data that RGB-Depth (RGB-D) systems collect and combine. Data acquisition, processing, analysis, and visualisation are some of the steps in this process, and each is a basic step for RGB-Depth and also essential to guaranteeing the accuracy as well as utility of the final dataset. A variety of applications in robotics, agriculture, augmented reality, and other fields are made possible by RGB-D systems, which depend on specialised hardware and software to carry out these duties effectively.

The first step in an RGB-D system's workflow is data acquisition. This entails concurrently recording depth information from a depth sensor and RGB pictures from a conventional camera. RGB cameras capture visual details like color and texture, while the depth sensor measures the distance between the sensor and objects in the environment. According to Pereira (2021) depth measurements are commonly obtained using one of three methods. Firstly, Time-of-Flight (ToF). Time of Flight (ToF) sensors are an active optical 3D acquisition method that measures the time it

takes for emitted light, usually infrared (IR) light, to travel from the sensor to the object and back. A more comprehensive 3D representation may be produced by combining this depth map, which is a 2D representation of the scene's 3D structure, with other data, such as RGB pictures from a different camera (Li, 2014).

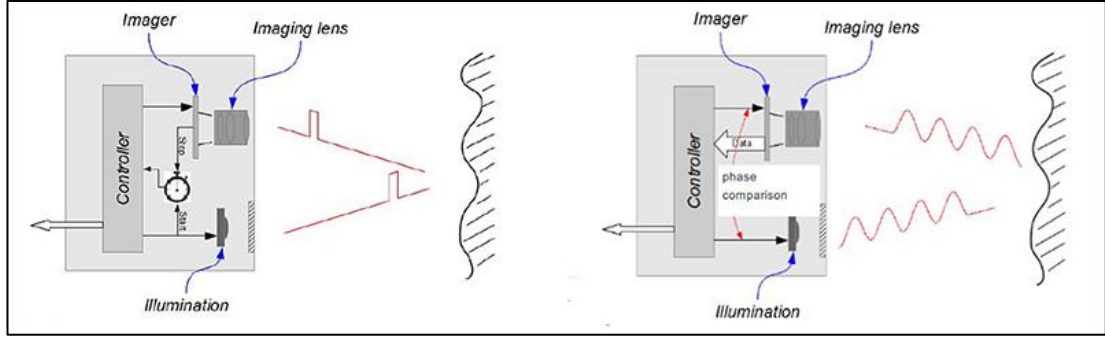


Figure 2.1: Operating principle of direct and indirect Time of Flight sensors
(Admin_Terabee, 2023).

Figure 2.1 shows the operating principle of direct and indirect Time of Flight sensors. Direct ToF sensors emit brief light pulses lasting only a few nanoseconds and determine distance by measuring how long it takes for the light to return after hitting an object. In contrast, Indirect ToF sensors use continuously modulated light and calculate distance by analyzing the phase shift in the reflected signal (Admin_Terabee, 2023).

Time-of-Flight (ToF) sensors have both advantages and disadvantages, making them suitable for specific applications. Among their advantages, ToF sensors excel in real-time performance by providing instant depth information, making them ideal for applications like robotics and augmented reality that require rapid and dynamic 3D

data acquisition, with some sensors operating at up to 60 fps. They also offer wider range capabilities, typically capturing distances up to 20 meters, which makes them particularly useful for navigation and object detection. However, ToF sensors have limitations. Their accuracy and resolution are generally lower compared to methods like structured light or stereo vision, which can be problematic for applications demanding detailed 3D models. Moreover, their performance can be affected by surface properties such as reflectivity, color, or transparency, leading to inaccurate depth measurements.

Despite these drawbacks, ToF sensors find widespread use in several fields. In robotics and automation, they assist with obstacle detection, navigation, and collision avoidance (Boston Dynamics, 2023). They are also used in gesture recognition and human-computer interaction, such as gaming consoles and smartphones, to provide interactive experiences.

Second method is Structured Light. An active optical 3D capture technique called "structured light" entails projecting a predetermined pattern typically a grid or a set of stripes onto the scene or object being scanned. A camera positioned at a known position and orientation with respect to the projector records the deformation of the projected pattern on the object's surface. Depth information may be extracted by the interaction of the projector, camera, and pattern deformation (Scharstein & Szeliski, 2003).

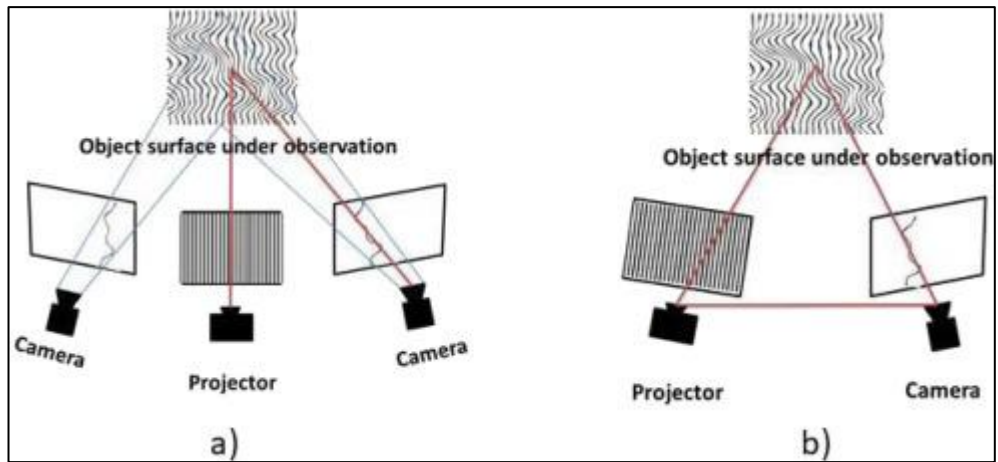


Figure 2.2: Illustration of the Structured Light operating principle. (Flores-Fuentes et al., 2022).

Figure 2.2 shows the classical configurations illustration of the Structured Light operating principle. **Figure 2.2 (a)** shows the system is based on two cameras and the use of one projector while **Figure 2.2 (b)** shows the system is based on one camera and the use of one projector.

Structured Light systems offer several advantages and disadvantages, making them effective for specific applications. One of their primary strengths is their high accuracy and resolution, allowing them to generate detailed 3D point clouds or depth maps suitable for precise measurements. Additionally, as an active method, structured light performs well under varying lighting conditions due to its reliance on its own light source. However, direct or bright indirect sunlight can interfere with the light projection, which means these systems are most effective in controlled indoor environments. On the downside, Structured Light systems have certain limitations. Their range is typically restricted, making them better suited for close-range scanning

or small objects, with an optimal working distance of 0.5–1.5 meters, though some can extend up to 3 meters. The performance can also be affected by the surface properties of the scanned object, such as high reflectivity or transparency, which may distort the projected patterns and introduce noise or artifacts into the 3D data. Despite these challenges, Structured Light systems are widely used in various fields. In manufacturing and industrial settings, they play a crucial role in quality control and inspection, ensuring precise measurements and identifying defects in components. According to Blz-Dental (2022), in dental and medical applications, structured light scanners are employed to produce accurate 3D representations of teeth, jaw structures, and body parts, which can be used for customized prosthetics and other medical purposes.

Last method is Stereo Vision, or stereoscopic vision. It is a passive 3D imaging technique that replicates how humans perceive depth. It employs two or more cameras, positioned a fixed distance apart (referred to as the baseline), to capture slightly different perspectives of the same scene. The resulting images, known as stereo pairs,

are analyzed using a stereo matching algorithm to locate corresponding features in both images.

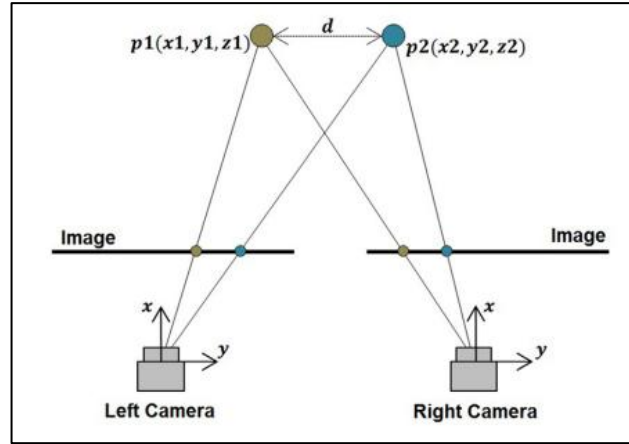


Figure 2.3: Illustration of the operating principle behind Stereo Vision

(Rodríguez-Quinonez et al., 2017).

This research from Rodríguez-Quinonez et al. (2017) says that measuring displacements and distances is only one of the numerous uses for stereo vision systems. They suggested the stereo vision system can follow two references in real time, as shown in **Figure 2.3**, allowing one to determine the spatial coordinates of both references. The displacement can then be tracked as a function of time using the formula for the distance between two points. Applications for displacement monitoring include stress analysis, occlusion detection, structural monitoring, geological monitoring, robot navigation, and more.

Stereo Vision systems offer several advantages that make them a popular choice for 3D depth acquisition in various applications. They are highly flexible, easy to set up, and capable of functioning effectively in a wide range of lighting conditions. Their ability to process depth information in real-time makes them ideal for tasks

requiring immediate feedback or rapid decision-making. Additionally, Stereo Vision systems are cost-effective compared to active methods like LiDAR or Structured Light, making them an appealing option for projects with budget constraints.

However, Stereo Vision systems also have notable limitations. They rely heavily on identifying corresponding features in multiple images, which can be challenging in scenes with low texture or repetitive patterns, leading to inaccurate or incomplete depth information. This issue, known as the stereo correspondence problem, is a common challenge. Furthermore, the method's accuracy can be affected by rapid changes in lighting or low-light conditions, as it depends on clear image capture. Stereo Vision also has a limited effective range, typically up to a few meters, and the accuracy of depth measurements decreases as the range increases. While active Stereo Vision techniques, which use additional light sources like lasers or structured light, can mitigate some of these challenges particularly in low-light environments they come with higher costs due to the added hardware.

Next step is Data Processing. Once the data is captured, it undergoes processing to refine and integrate the RGB and depth streams. In order to adjust for distortions and inconsistencies brought on by the physical offset between the RGB camera and depth sensor, this step entails matching the RGB picture and depth map. Filters and algorithms are applied to reduce noise in the depth data, such as artefacts from reflecting or clear surfaces, and calibration procedures are utilised to guarantee precise alignment (Jin et al., 2021).

The third step in RGB-Depth data processing is data analysis. This involves using the depth and color information to extract meaningful insights, such as identifying shapes, textures, and object locations. In agriculture and other fields, this step is important for tasks like object recognition and decision-making. Feature extraction is a key part of the process, where relevant details from the RGB and depth data are identified. The depth information adds useful 3D context, helping to distinguish between objects that may look similar in color but differ in size or distance. Advanced models, such as CIR-Net, improve accuracy by combining RGB and depth features more effectively using attention-based methods (Cong et al., 2022).

After feature extraction, machine learning models such as convolutional neural networks (CNN) are commonly used to analyze RGB-D data. These models help identify patterns for tasks like object detection and segmentation. Some networks, like SP-Net, are designed to combine both RGB and depth features effectively while maintaining the strengths of each data type (Zhou et al., 2021).

Accurate 3D geometric modelling and analysis are made possible by depth data. Reconstructing 3D environments is made possible by methods like point cloud generation and voxel-based representations, which make jobs like virtual reality and robotic navigation easier. According to Kulkarni et al. (2023) recent developments, such implicit function modelling, show applications in high-fidelity reconstruction and scene comprehension by explicitly predicting complex 3D structures from pictures using RGB-D data. Yin et al. (2023) studied that by merging RGB and depth data to provide rich, transferable feature representations, pretrained models like DFormer

improve semantic segmentation and increase performance in tasks like object identification and saliency prediction.

The final step in RGB-D data processing is visualization. At this stage, the data is displayed in formats such as 3D point clouds or textured models, combining depth for structure and RGB for color details. These visualizations are useful in applications like virtual reality and agriculture, where accurate 3D models help in decision-making. Tools also allow users to interact with the data, such as measuring dimensions or analyzing object positions. In robotics, visualization helps machines navigate and understand their surroundings in real time.

2.3.3 Applications of RGB-Depth Data in Agriculture

One of the example applications of RGB-D technology in agriculture is in crop phenotyping, a process focused on analyzing plant traits such as height, canopy volume, and biomass. The RGB component captures detailed color and texture information, while the depth data provides a 3D perspective of the plant structure. Together, these features enable comprehensive and accurate assessments of plant attributes. For example, sensors like the Intel D435 have been proven effective for close-range phenotyping tasks in various lighting conditions. These sensors reliably measure critical plant characteristics, such as height and volume, which are essential for improving irrigation and fertilization strategies (Vit & Shani, 2018). Chéné et al.(2012) demonstrated the potential and capability of RGB-D imaging systems for 3D measurements in the context of plant phenotyping. They demonstrated how

MICROSOFT KINECT I can resolve individual leaves, enabling automated leaf orientation measurement in indoor settings. RGB-D technology has been used in a variety of plant phenotyping tasks, showcasing its versatility in both indoor and outdoor conditions. For example, Xia et al.(2015) developed an algorithm for segmenting leaves using RGB-D data and demonstrated its effectiveness with greenhouse-grown paprika leaves. Similarly, Wang and Li (2014) showed that they have used RGB-D cameras to estimate the size of sweet onions in indoor settings and other researchers also use it to measure the size of mango fruits directly on trees in the field Z. Wang et al. (2017) . Other studies have focused on broader canopy structure measurements. For instance Hui et al. (2018) showed that they used RGB-D cameras to analyze the canopy structure of small plants in the field, while Azzari et al. (2013) utilized the Microsoft Kinect I sensor to assess vegetation structure. In another application, Jiang et al. (2018) developed an algorithm to measure cotton canopy size in field conditions. They found that multidimensional and multivariate traits derived from 3D imaging provided better yield predictions than traditional univariate traits, highlighting the advantages of 3D imaging techniques. Sa et al. (2017) presented that RGB-D technology has also been employed for more specific tasks, which used a short-range Intel RealSense F200 depth camera to develop a 3D visual detection method for identifying sweet pepper peduncles in the field. For example, the WE3DS RGB-D dataset introduced semantic segmentation benchmarks to differentiate between soil, seven crop species, and ten weed species. By combining depth data with RGB imagery, model performance improved, achieving up to 70.7% mean Intersection over Union (mIoU) in segmentation tasks (Kitzler et al., 2023). These examples

illustrate the significant potential of RGB-D technology for improving the precision and efficiency of plant phenotyping across various applications.

RGB-D sensors also play a key role in early disease detection by merging the visual details captured by RGB imagery with the spatial perspective provided by depth data. This combination enables the identification of abnormalities in plant morphology. For example Skoczeń et al. (2021) studied that RGB-D cameras have been used in greenhouses to monitor crops for signs of disease. Structural changes, such as distorted leaves or unusual growth patterns, are identified through depth data, while discoloration is analyzed using the RGB channels.

2.3.4 Applications of RGB-Depth Data in Oil Palm

One key application of RGB-D technology in oil palm farming is ripeness detection, a crucial process that directly impacts oil yield and quality. Traditionally performed manually, this task is now being automated with the help of RGB-D systems. These systems combine RGB color imaging with depth data to assess the ripeness of oil palm fruit bunches. For instance, RGB features like color intensity and texture are used to determine ripeness stages, while classification is performed using techniques such as fuzzy logic or artificial neural networks (ANNs). An example of this application is a mobile app developed with an ANN trained on RGB color data. The app classifies palm oil bunches into three categories: ripe, underripe, and overripe. Using back propagation techniques for data processing, the system achieved an impressive classification accuracy of 96% (Alzahati & Mohamad@Hamza, 2016). To

predict yields, data from these features is often analyzed using linear regression models or advanced machine learning algorithms. For example, a system combining hyperspectral imaging with RGB color imaging was developed to estimate the weight and ripeness of oil palm fruit bunches. This approach achieved a coefficient of determination (R^2) of 0.989 for weight estimation, highlighting its potential for highly accurate yield predictions (Junkwon et al., 2009).

Early detection of diseases in oil palm plantations is essential to reduce yield losses and effectively manage pests. RGB-D technology significantly enhances disease monitoring by identifying subtle changes in plant structure and color caused by infections. This method combines RGB imagery, which highlights visible disease symptoms such as yellowing or spots, with depth data that detects structural anomalies in palm fronds or trunks. Machine learning models trained on annotated RGB-D datasets are then used to classify plants as healthy or diseased (May & Amaran, 2011). For instance, a computer vision-based system utilizing RGB-D data was developed to detect Ganoderma, a fungal disease that poses a serious threat to oil palms. By analyzing both structural deformations and visible symptoms, the system improved early detection rates, allowing for targeted interventions and reducing economic losses (Ismail & Razali, 2010). This approach underscores the potential of RGB-D systems in safeguarding plantation health and productivity.

2.4 Point Cloud Data

2.4.1 Introduction about Point Cloud Data

Point cloud data is a set of points in a 3D coordinate system used to represent the shape and surface of objects or environments. It is typically generated using technologies like LiDAR, photogrammetry, or laser scanning (Yang et al., 2023). Originally developed for military and surveying use in the mid-20th century, point cloud data has become more widely used over the past two decades due to advances in 3D sensing. The integration of deep learning and artificial intelligence has further improved point cloud processing, enabling applications such as 3D object detection and autonomous navigation (Bello et al., 2020).

3D data can be represented in several formats, such as depth images, point clouds, meshes, and volumetric grids. Among these, point clouds are widely used because they preserve the full geometric details in three dimensions without simplification. This makes them suitable for applications like scene interpretation in robotics and autonomous driving. While deep learning has advanced in many fields, Ronneberger et al. (2015) studied that applying it to 3D point clouds remains challenging due to their unstructured format, high dimensionality, and limited dataset availability. According to Guo et al. (2020), Particularly in the past five years, deep learning on point clouds has gained increasing interest. A number of publicly accessible datasets are also made available, including the KITTI Vision Benchmark Suite, ModelNet, ScanObjectNN, ShapeNet, PartNet, S3DIS, ScanNet, Semantic3D, and ApolloCar3D. With more and more techniques being proposed to address different point cloud processing issues, such as 3-D shape classification, 3-D object detection

and tracking, 3-D point cloud segmentation, 3-D point cloud registration, 6-DOF pose estimation, and 3-D reconstruction, these datasets have further accelerated the study of deep learning on 3D point clouds. A taxonomy of existing deep learning methods for 3D point clouds is shown in **Figure 2.4** below.

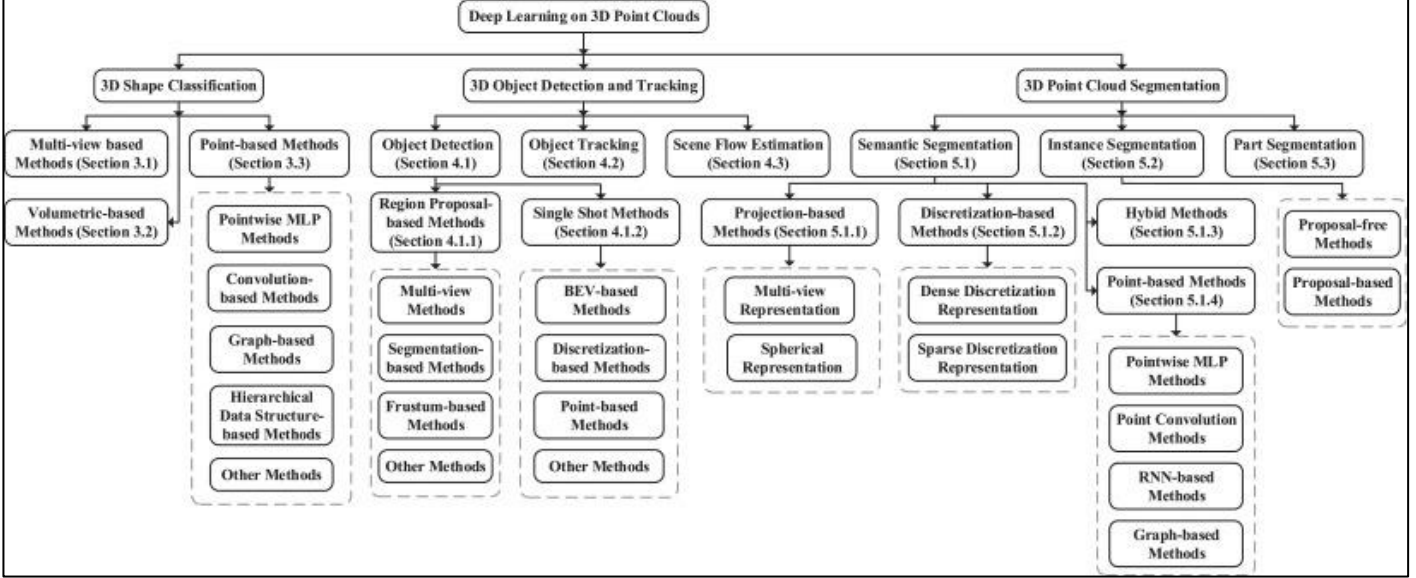


Figure 2.4: A taxonomy of deep learning methods for 3D point clouds (Guo et al., 2020)

Point cloud data is a useful way to represent 3D objects and spaces. Each point has x, y, and z coordinates, and may include additional information like color or intensity. Metadata such as timestamps and scanner details adds context to the data. Point clouds can be created using technologies like LiDAR, photogrammetry, and 3D scanners. Wang & Kim. (2019) presented that drones are also commonly used for large-area surveys in agriculture and environmental monitoring. Point clouds can be structured, with points arranged in a grid, or unstructured, with randomly placed points suited for complex or moving scenes. They are stored in various formats like LAS/LAZ for LiDAR, PLY for color and surface data, E57 for multi-scanner systems,

and XYZ for basic coordinate data (Cura et al., 2017). Point clouds are widely used across industries, including construction and cultural heritage preservation, where they help capture detailed 3D models of buildings and artifacts for future reference (Yang et al., 2023).

2.4.2 Characteristic about Point Cloud Data

Point cloud data is valued for its precision and ability to capture detailed 3D shapes, making it useful in areas like architectural modeling and quality inspection. It is also highly scalable, suitable for both small objects and large-scale environments like entire cities. With modern tools such as Apache Spark, even large point cloud datasets can be processed efficiently. In addition to geometric accuracy, point clouds provide multidimensional data attributes such as color, reflectance, and temporal information. These attributes enrich the dataset, enabling advanced applications like material analysis and 3D object recognition (Cheng et al., 2023). However Liu et al. (2019) investigated that, their unstructured nature sets them apart from other data types, such as images. While this irregularity presents challenges for processing, it offers flexibility in representing complex geometries.

Point cloud data often comes in large volumes, requiring efficient storage and processing. Techniques like clustering and deep learning-based compression help manage storage while keeping important details (Sun et al., 2019). Moreover, Wang & Kim. (2019) studied that point clouds offer valuable spatial and temporal information, useful for tasks like monitoring construction progress and detecting

environmental changes. Their compatibility with machine learning allows for tasks such as classification and segmentation. Methods like dynamic graph convolution are especially effective for analyzing complex data (Wang et al., 2018). However, due to their size and density, point clouds can be difficult to visualize, often requiring specialized tools like PCL and MeshLab (Pajic et al., 2018).

Finally, recent advancements have enabled real-time processing capabilities, a critical feature for applications in dynamic environments such as autonomous driving and robotics. Efficient clustering algorithms, such as those based on Euclidean distance, ensure low-latency data handling in these scenarios (Najdataei et al., 2018).

2.4.3 Process of Point Cloud Data

Point cloud data provides an accurate way to represent 3D shapes and environments. It starts with data collection using tools like LiDAR, photogrammetry, or structured light scanners, which capture millions of points on an object's surface. Depending on the use case, either ground-based scanners or drones are used for precise data gathering in different settings (Wang et al., 2019).

Once collected, raw point cloud data goes through several processing steps to improve its quality. Filtering methods like statistical outlier removal and radius filtering help reduce noise from the environment or scanning errors. Huang et al. (2024) presented that combining these methods results in cleaner and more reliable data. Registration techniques, such as Iterative Closest Point (ICP), are then used to align multiple scans into one coordinate system, especially useful in large or complex

projects. Segmentation divides the data into meaningful parts, like individual objects, using methods such as mathematical morphology to maintain spatial accuracy (Balado et al., 2020). Finally, Lee et al. (2021) shows that compression techniques, like voxel-based compression, help reduce data size while preserving important details. Visualization and analysis make point cloud data more useful. Tools like Open3D and PCL allow users to explore 3D data through actions like zooming and rotating. Techniques such as gridification help simplify the data for tasks like segmentation. For analysis, deep learning models are used to detect, classify, and extract features from 3D data, taking advantage of the point cloud's unstructured nature (Ding et al., 2023).

In advanced applications like autonomous driving, real-time processing of point cloud data is crucial. Techniques such as voxel grid filtering help reduce data size while keeping accuracy. Semantic analysis further improves understanding by labeling parts of the point cloud, useful in tasks like building or terrain mapping. Models like SnowflakeNet enhance this process with more detailed analysis.

2.4.4 Applications of Point Cloud Data in Agriculture

Point cloud data is increasingly used in agriculture to create accurate 3D models of crops and farmland. Collected through tools like LiDAR or RGB-D cameras, this data helps map field topography, monitor crop growth, and identify problem areas. The study from Zhao et al. (2024) showed that drones equipped with point cloud sensors can generate detailed crop maps, allowing farmers to assess plant health and manage resources more effectively.

Point cloud data also supports the development of agricultural digital twins virtual models of crops and environments used to simulate farming activities and predict outcomes like yield. Catala-Roman et al. (2024) showed that in orange groves, point clouds from photogrammetry help test different management strategies. Xu et al. (2023) investigated that point cloud completion techniques using models like PointNet and EdgeConv can fill gaps caused by leaf occlusion, enabling full 3D tree models for analyzing growth and estimating production.

Lastly, point cloud data integration has improved cloud-based agricultural systems. Better crop health management and resource allocation are made possible by these systems' real-time monitoring and data sharing capabilities. For example Junaid et al. (2021) studied that cloud platforms that integrate AI algorithms with IoT sensors have been utilized to analyse various agricultural data, increasing sustainability and production.

2.4.5 Applications of Point Cloud Data in Oil Palm

Oil palm plantations require accurate monitoring and management due to their large scale. Point cloud data has improved efficiency and yield estimation by enabling detailed 3D mapping and analysis. Using UAVs with Structure-from-Motion (SfM) photogrammetry, plantations can be mapped with high accuracy, allowing for automatic palm identification and segmentation. Fawcett et al. (2019) reported 98.2% accuracy in identifying individual palms using local height maxima in dense plantations in Sarawak, Malaysia. Moreover, metrics such as canopy and stem height

derived from point clouds have been validated against ground data, achieving root mean square error (RMSE) values of 0.27 m and 0.45 m for palms aged 7 and 10 years, respectively. This capability aids in monitoring growth stages and estimating biomass effectively.

Fawcett et al. (2019) also studied that carbon accounting and biomass estimates are two more crucial applications in addition to mapping. Point cloud data makes it easier to calculate per-palm biomass by examining canopy structure and stem dimensions. These estimates are crucial for evaluating sustainability and environmental effect. Carbon accounting initiatives in oil palm plantations have benefited from the successful use of UAV-based point cloud techniques for measuring above-ground biomass at the tree level.

2.5 Weight Estimation in Agriculture

2.5.1 Introduction of Weight Estimation in Agriculture

Weight estimation plays a key role in improving productivity and efficiency in areas like crop management, post-harvest processing, and livestock farming. Technologies such as AI, machine learning, and image processing are used to estimate weight without direct contact. This reduces manual labor, lowers handling stress, and helps optimize resource use. Weight estimate techniques have been transformed by recent technological developments. For example, the very accurate weight estimation of lambs using 3D imaging has improved operating efficiency and decreased animal discomfort. Similarly, to enable automated sorting and grading in agricultural

processing, machine vision and neural networks have been used to estimate the weight of crops like tomatoes and mangoes (Nyalala et al., 2021). In livestock farming, weight estimation using methods like linear regression, random forest models, and fuzzy logic has provided farmers with tools to monitor growth, optimize feed usage, and improve economic outcomes (Setiawan et al., 2024). These methods integrate data-driven approaches and traditional agricultural knowledge to deliver precise and actionable insights. Weight estimate helps with effective sorting and packing after harvest. To determine the weight and volume of fruit with irregular shapes, such as avocados and sweet potatoes, vision-based devices have been created. These non-invasive techniques have shown themselves to be precise and economical, greatly aiding in export quality control and market planning (Huynh et al., 2022).

2.5.2 Method of Weight Estimation in Agriculture sector

In agriculture, weight estimation methods are important for measuring crops and livestock. While manual methods are simple and suitable for small-scale use, they can be labor intensive and less accurate. In contrast, advanced methods such as AI and vision-based technologies offer higher accuracy and efficiency, making them more suitable for large-scale or automated operations.

The studied from Liu et al. (2024) developed a non-contact method to estimate pig weight and body dimensions using 3D point cloud data captured from a depth camera placed above the animals. Data from 258 pigs was processed using voxel filtering and K-means clustering to isolate the back region, with posture correction

applied before extracting body widths. A convolutional neural network with a multi-head attention mechanism was used for weight prediction, combining 3D and RGB data. The method achieved measurement errors below 4% and a weight prediction RMSE of 11.552 kg, with fast processing and minimal stress to the animals. In addition, Paudel et al. (2023) studied PointNet, a deep learning model, to predict pig weights from 3D point clouds and compared it to traditional volume-based regression. Using data from 249 pigs captured with a stereo depth camera, PointNet achieved an R^2 of 0.94 and RMSE of 6.87 kg, outperforming the volume-based method. It performed especially well for pigs under 55 kg and was more robust to posture and movement, making it suitable for automated on-farm applications.

The studied Aung et al. (2020) developed a low-cost method to estimate the volume and weight of mangoes using a single top-view image from an RGB camera. The process involves image segmentation, curve fitting, and estimating volume through the sum of transverse elliptic cylinders, with thickness predicted using linear regression. The method achieved 95% accuracy for both volume and weight estimation with a fast processing time of 0.2 seconds per mango. In other research, Mohd Fauzi et al. (2022) proposed a 2D contactless computer vision approach for estimating the volume and weight of apples. Using images from two smartphones and Mask R-CNN for segmentation, the volume was estimated using a modified sphere formula and pixel-to-centimeter calibration. The method achieved estimation errors of 11.97% for volume and 11.49% for weight, showing that 2D image-based systems can be practical for industrial applications, with further improvements possible through 3D techniques.

The researchers demonstrated how often automated and effective image processing techniques were for weight and volume estimation in the agriculture sector. Chang et al. (2017) proposed a method for estimating object volume using 3D point cloud data. The technique involves slicing the point cloud along the z-axis, further dividing each slice along the x and y-axes, and identifying boundary points. Polynomial curves are fitted to these points, and the area of each slice is calculated and summed to get the total volume. Tested on various shapes, the method showed high accuracy with errors below 0.5%. It was implemented in C++ using the Point Cloud Library and depends on the quality of the input data.

2.6 Summary

This chapter reviewed key topics related to estimating the mass of oil palm Fresh Fruit Bunches (FFB), including the oil palm industry, RGB-Depth (RGB-D) data, point cloud data, and weight estimation methods in agriculture. The review supports the need for an automated approach to estimate FFB mass, as traditional weighing methods are slow and can delay harvesting and transport. Technologies like RGB-D and point cloud data offer faster and more consistent alternatives, improving efficiency in plantations. RGB-D data combines color and depth information to generate detailed object views. It is used in agriculture to measure plant height, leaf size, and fruit volume. Devices like Intel RealSense and Microsoft Kinect are commonly used to collect this data, which reduces manual work and provides real-time 3D capture. In oil palm applications, RGB-D has been used to detect fruit ripeness and predict yield using image features and machine learning. These studies show that RGB-D is suitable for

FFB volume estimation. Point cloud data, made up of 3D points representing objects or surfaces, is widely used in agriculture. Collected via depth cameras, LiDAR, or drones, it helps create 3D models of trees and crops for measuring canopy structure, tree height, and tracking growth. In oil palm plantations, it is used to detect individual palms, estimate biomass, and monitor health. Point cloud data also supports automation through drones and robots, making it useful for estimating FFB size and volume. The final section discussed weight estimation methods in agriculture. Many studies have used RGB images, 3D imaging, and point clouds to estimate weight. Simpler approaches using top-view images to estimate fruit size and volume have also performed well. These examples demonstrate that combining volume estimation with known or predicted density can effectively estimate weight without manual weighing.

CHAPTER 3

METHODOLOGY

3.1 Overview

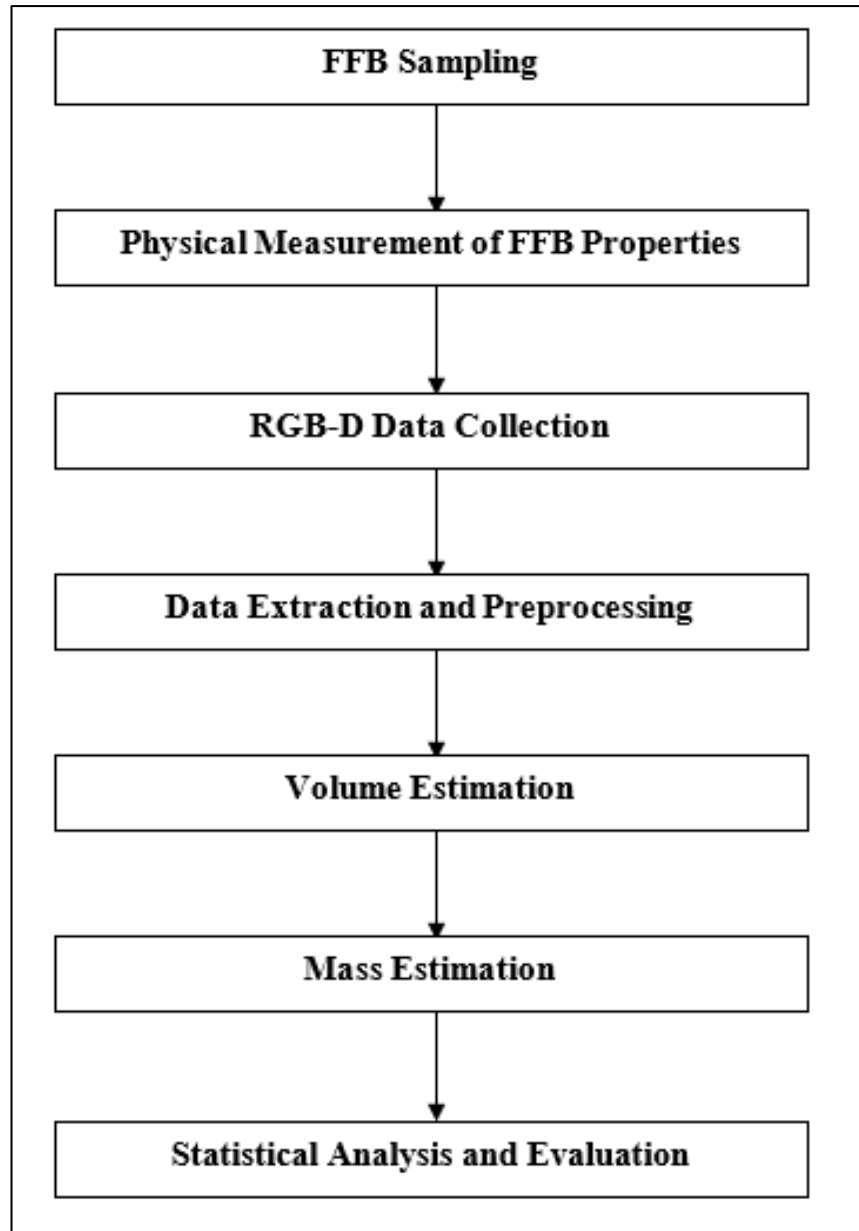


Figure 3.1: Workflow of FFB Mass Estimation Using RGB-D Data

The study was carried out at Universiti Putra Malaysia (UPM). **Figure 3.1** shows the flowchart process began with FFB sampling, where 50 freshly harvested fruit bunches were selected from Havys Oil Mill Sdn. Bhd. This step provided the samples needed for both objectives. The next step was the physical measurement of FFB properties, where actual mass was measured using a mechanical spring scale, and actual volume was obtained using the water displacement method. These measurements were used to calculate the actual density of each fruit bunch, which directly addressed Objective 1, to determine the density of oil palm Fresh Fruit Bunch (FFB). Following that, RGB-D data collection was carried out using the Intel RealSense D435i camera mounted in a top-down position. This step provided both color and depth data for each FFB. The depth information was used to generate point clouds for volume estimation, while the color information supported segmentation of the fruit from the background. The captured depth data was then processed to extract and clean the point cloud of each FFB, preparing it for volume computation. Volume estimation was performed using CloudCompare software, where a 2.5D volume estimation method was applied to each point cloud. This provided the estimated volume of each FFB and directly addressed Objective 2, which to determine the variations between volume estimated using RGBD sensing data and physically measured volume of oil palm FFB. After that, mass estimation was carried out by multiplying the estimated volume with the average actual density, linking results from both objectives. Lastly, statistical analysis and evaluation were performed to compare the estimated values with the actual measurements. Correlation analysis and error evaluation were used to determine the relationship between estimated and actual data. This final step provided insight into the accuracy

and reliability of the RGB-D-based estimation method, and helped validate both objectives of the study.

3.2 Collecting and Processing the Physical Data for Mass Estimation



Figure 3.2: Study location at Havys Oil Mill Sdn. Bhd., Semenyih, Selangor.

3.2.1 FFB Sampling

Figure 3.2 shows that this study was conducted at the loading ramp of Havys Oil Mill Sdn. Bhd., located in Semenyih, Selangor (2.9661° N, 101.8496° E). A total of 50 fresh fruit bunch (FFB) samples of oil palm were selected for data collection. The duration of data collection about 3 month. The oil palm FFB came from many source oil palm plantations nearby. Usually, trucks arrive at the loading ramp daily to deliver freshly harvested oil palm fresh fruit bunches (FFB). After 6:00 PM, the ramp is

closed. After this time, all of the FFB that is delivered to the loading ramp for that day is collected in trailer trucks and transported to the palm oil mill for processing the following day. Therefore, the sample collected came from freshly harvested oil palm fruits, not from old or previously stored bunches.

3.2.2 Physical Measurement of FFB Properties

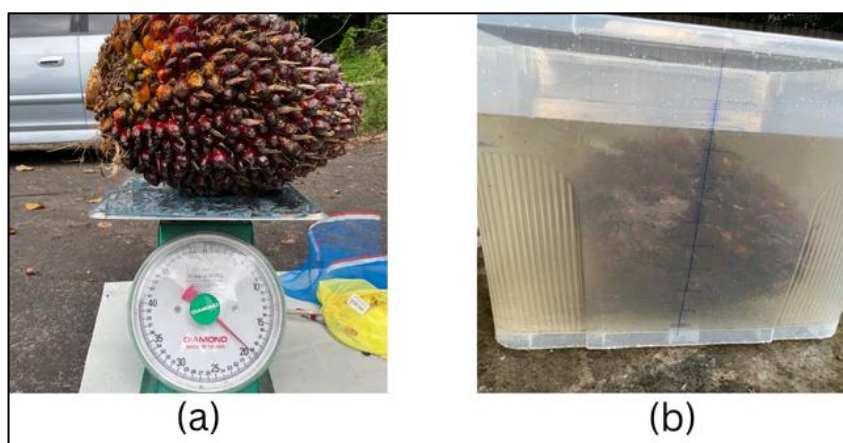


Figure 3.3: Process of physical measurement of FFB Properties **(a)** Mass measurement of oil palm FFB using mechanical spring scale, **(b)** Volume measurement using water displacement method in a custom-built transparent

Following video collection, physical measurements of the FFB samples were performed. Each sample was allowed to rest naturally on a flat surface, and its dimensions (length, width, and thickness) were recorded using a Class II measuring tape (Stanley, USA), accurate to ± 0.5 mm at 1 meter. These measurements were recorded in centimeters and stored in an Excel sheet.

For mass measurement, a mechanical spring scale (DIAMOND, Taiwan, Model: TH02FS) with a 50 kg capacity and 0.2 g precision was used (**Figure 3.3 (a)**).

The masses of the FFB samples ranged between 8 kg to 25 kg. To measure the volume, a water displacement method (WDM) was applied. A custom-built setup using a thickened transparent storage box (110-liter capacity) (Dimensions: Internal: 58 x 40 x 37.5 cm , External: 63.5 x 44 x 40 cm) marked at 1-liter intervals was used as an oil palm FFB volume measurement system as shown in **Figure 3.3 (b)**. The 1-liter intervals was determined by each water that was poured into that storage box using a measuring cylinder (500 ml) from the lab. The box was filled to an initial water level of 70 liters, and each FFB was submerged to calculate its volume by recording the final water level. The actual volume was calculated as the difference between the initial and final water levels. The density was then determined by dividing the mass of an oil palm FFB by the measured volume. Each time after measuring the volume of an oil palm fresh fruit bunch (FFB), its mass is recorded again to determine whether water exposure during the volume measurement process affects the FFB's mass or not. The average of each measurement was noted and tabulated (**Appendix A**).

3.3 Collecting and Processing the RGB-D Data for Volume Estimation

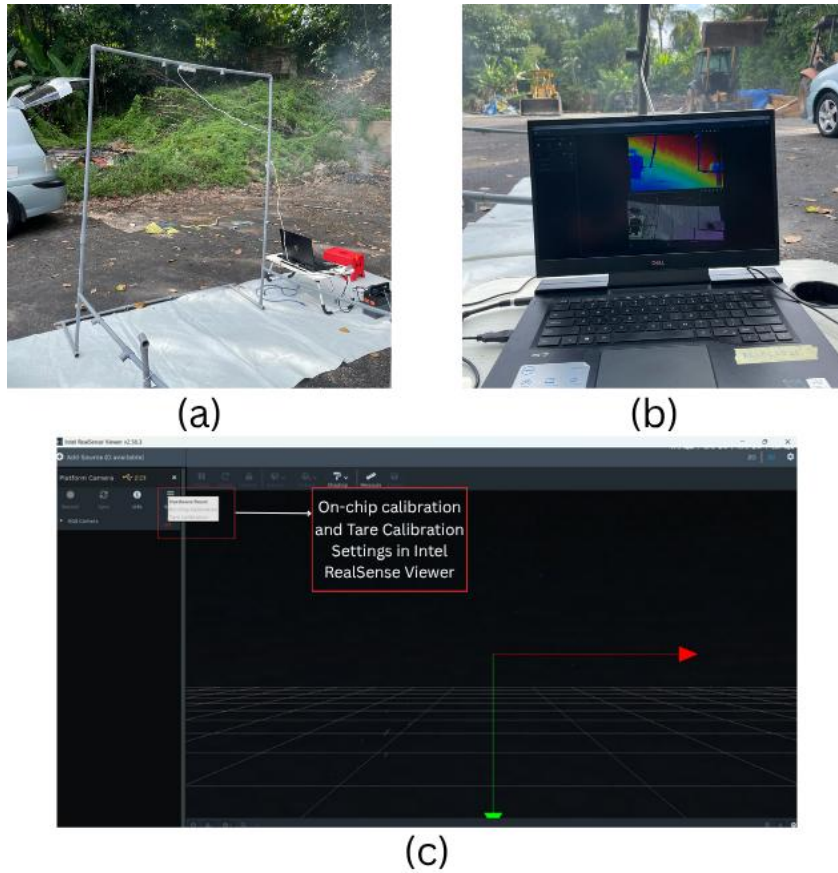


Figure 3.4: Overview of RGB-D Data Collection; (a) Intel RealSense D435i RGB-D camera mounted at 1.5 m height for video acquisition, (b) RGB and depth data recording via laptop using Intel RealSense Viewer. (c) On-chip self-calibration and tare calibration in Intel Realsense Viewer.

3.3.1 RGB-D Data Collection

The Intel RealSense D435i camera used in the experiment was positioned at a height of 1.5 m from the ground as shown in **Figure 3.4(a)**, and was connected to a laptop via a USB 3.0 cable with screw locks and a 1 m long. Manual triggering of video

acquisition was done using Intel RealSense Viewer version 2.56.3 (**Figure 3.4 (b)**). Video captures of each FFB were conducted at 30 frames per second (fps) for both RGB and depth video. The RGB videos were recorded at a resolution of 1280×720 pixels with a $69.4^\circ \times 42.5^\circ$ field of view (FOV), while the depth videos were recorded at 848×480 resolution with an $86^\circ \times 57^\circ$ FOV. The camera was calibrated in Intel RealSense Viewer software version 2.56.3 which is on-chip self-calibration and tare calibration as shown in **Figure 3.4 (c)**. When taking the oil palm FFB's RGB and depth video, each FFB was rotated in four orientations, with approximately 7 seconds of video captured per orientations to ensure complete surface coverage. All videos recorded will save in .bag file format. The .bag file refers to the rosbag file that Intel RealSense Viewer saves in. The file format used by ROS (Robot Operating System) to store recorded sensor data. The .bag file is used to record sensor streams from an Intel RealSense camera and it can playback those recordings later for our testing, analysis, or development without needing a live camera. A total of 50 RGB video and 50 depth video had been captured respectively to the total sample of oil palm FFB which is 50 oil palm FFB. The video was taken in bright and good weather.

3.3.2 Data Extraction and Preprocessing

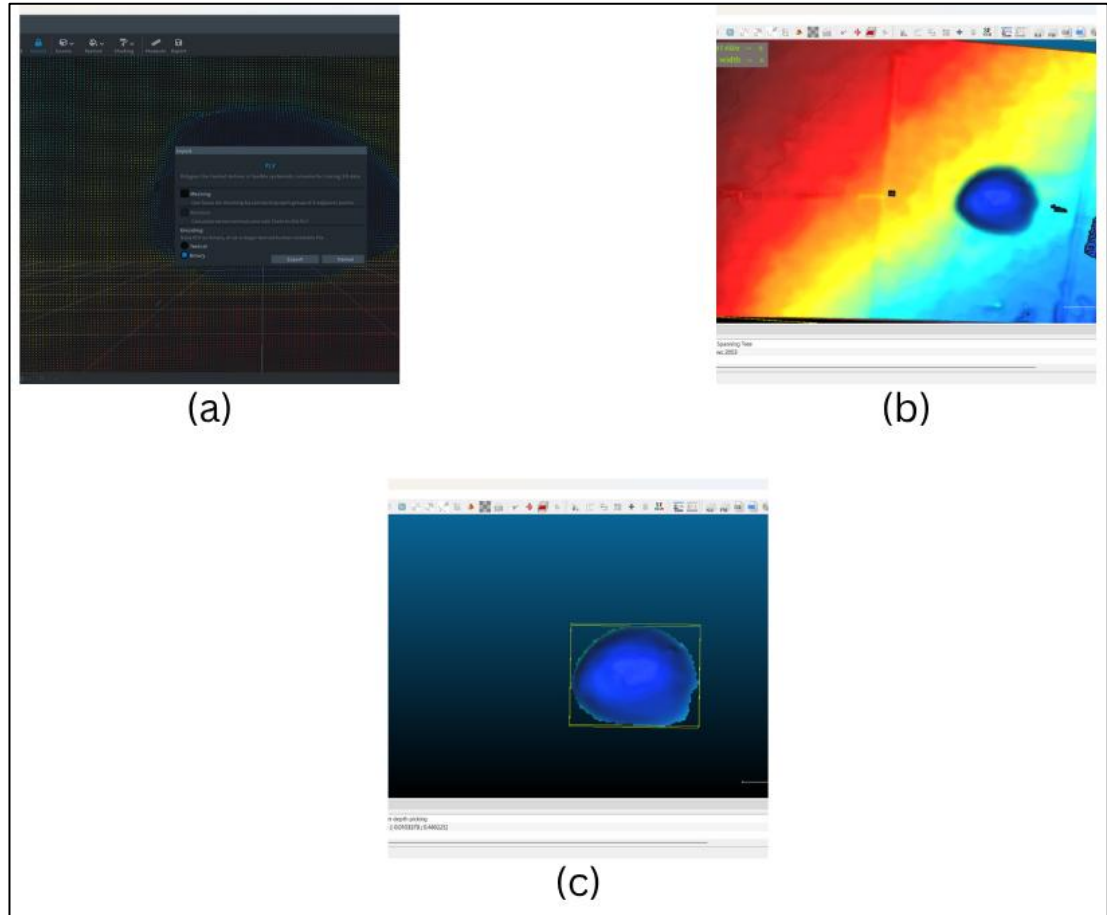


Figure 3.5: Data extraction and preprocessing of oil palm FFB point clouds; **(a)** Exporting point cloud data from depth video as .ply format using Intel RealSense Viewer, **(b)** Computing normal vectors and applying 2.5D Delaunay meshing in CloudCompare, **(c)** Segmentation and filtering to isolate the FFB from background noise for volume analysis.

Point clouds were extracted from the depth videos using Intel RealSense Viewer software version 2.56.3. After loading the depth video .bag files, the recorded depth video will appear in Intel Realsense Viewer software and to extract the point clouds

from this depth video, it needs to be exported in binary (.ply) format as shown in **Figure 3.5 (a)**. These oil palm FFB's point clouds were processed in CloudCompare v2.14.alpha software, where normal vectors were computed to determine surface orientation, and Delaunay 2.5D meshing (on the XY plane) was applied (**Figure 3.5 (b)**) to approximate surface geometry of the object by projecting the 3D points onto a 2D plane and connecting them, enabling further analysis such as volume estimation.

Filtering and segmentation were applied to remove unwanted objects and noise. The segmentation process involved isolating the FFB from the background using manual selection tools in the software (**Figure 3.5 (c)**). This ensured only the relevant surface data of the fruit was retained for volume analysis.

3.3.3 Volume Estimation

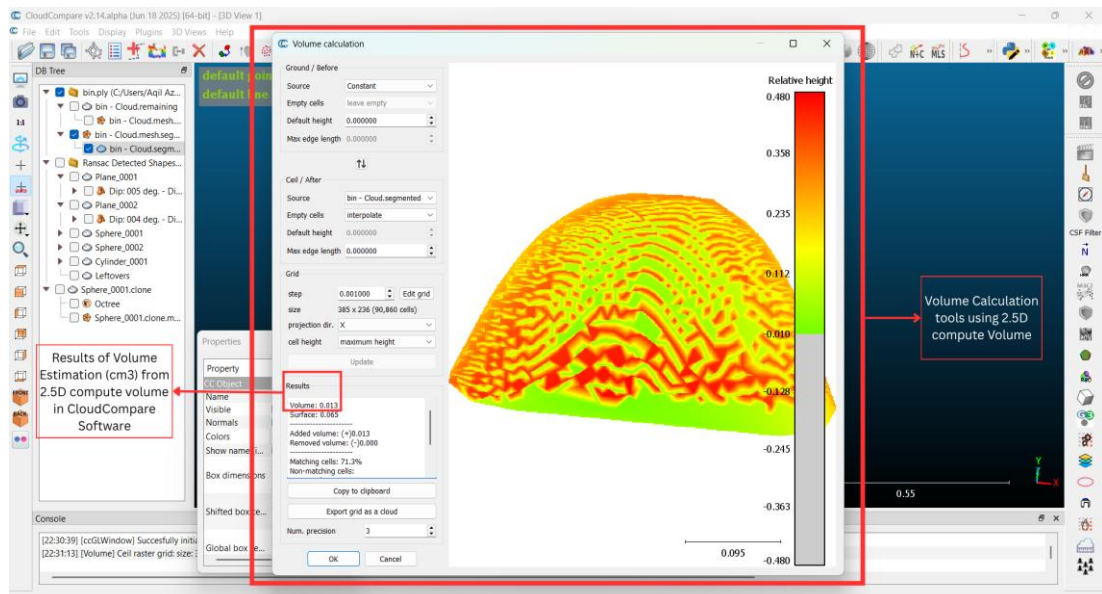


Figure 3.6: Estimate the volume of the oil palm FFB point cloud, the 2.5D Volume Calculation tool in CloudCompare

To estimate the volume of the oil palm FFB point cloud, the 2.5D Volume Calculation tool in CloudCompare v2.14.alpha software was used as showcased in **Figure 3.6**. A constant flat surface was set as the ground reference, while the oil palm FFB's point cloud segmented mesh was used as the upper surface. Empty cells in the mesh were interpolated to close gaps for more accurate volume calculation. The projection direction was set along the X-axis. A 0.001 grid step size was used to define the resolution of the calculation. The cell height was defined using the maximum point height within each grid cell. Volume estimation results were generated automatically upon running the calculation. From volume estimation results, mass estimation can be defined by simply multiplying volume estimation with average density of actual oil palm FFB.

3.4 Statistical Analysis and Evaluation of the result

To evaluate the Collecting and Processing the Physical Data for Mass Estimation and RGB-D Data for Volume Estimation for this study, two types of statistical analysis were carried out by descriptive analysis and correlation analysis. These methods were used to summarize the characteristics of the data and to assess the strength of relationships between the actual and estimated values of volume and mass. All calculations were based on standard statistical formulas.

3.4.1 Descriptive Analysis

Descriptive analysis was used to summarize and understand the overall pattern of the data collected, including actual mass, size, volume, and density of FFB. The following statistical parameters were calculated using Mean, Median, Interquartile Range (IQR) and Skewness as shown in Equations (1), (2), and (3).

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Where $\bar{x}$, mean (average), x_i is each individual data value and n total number of data points. Mean is use for calculates the average value of a dataset by summing all the values and dividing by the number of data points, n.

Second is Median. The median is the middle value in the sorted dataset. If there is an even number of data points, the median is the average of the two middle values. It is useful for identifying the central value, especially when the data has skewness or outliers.

$$IQR = Q_3 - Q_1 \quad (2)$$

Where Q_1 is first quartile (25th percentile) and Q_3 is third quartile (75th percentile). IQR measures the range within which the middle 50% of the data lies. It is useful for understanding the spread of the central data values.

$$Skewness = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3 \quad (3)$$

Where n is total number of data points, $\bar{x}$ mean of the data, x_i is each individual data value and s is standard deviation. Skewness describes the asymmetry of the data. A skewness value close to 0 suggests a symmetrical distribution. Positive skewness means the right tail is longer, while negative skewness means the left tail is longer.

3.4.2 Correlation Analysis

Correlation analysis was used to determine the strength and direction of the linear relationship between two variables, specifically between Actual Volume versus Estimated Volume and Actual Mass versus Estimated Mass. The equation that are used in correlation analysis are Pearson Correlation Coefficient (r) and Coefficient of Determination (r^2) as shown in equation (4) and (5). For validation part the equation that was used basically equation that related to mass, volume and density as shown in equation (6).

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \quad (4)$$

Where x_i is actual data values (actual volume or mass), y_i is estimated data values (estimated volume or mass), $\bar{x}$ is mean of actual data, $\bar{y}$ is mean of estimated data and n is number of data points. This formula measures how strongly two variables are linearly related. The value of r ranges from -1 to 1. A value near 1 indicates a strong positive relationship, near -1 indicates a strong negative relationship, and near 0 means no linear correlation.

$$\text{Coefficient of Determination, } r^2 = (r)^2 \quad (5)$$

This value shows how much of the variation in the actual values can be explained by the estimated values. For example, if $r^2 = 0.8119$, then 81.19% of the variation in actual mass can be predicted by the estimated mass.

$$\text{Estimated Mass, } m = p \times v \quad (6)$$

Where p is average actual density of 50 oil palm FFB samples (in kg/m³) and v is estimated volume of FFB (in m³). This formula was used to calculate the estimated mass of each FFB using the estimated volume from CloudCompare and the average actual density from physical measurements.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

This chapter presents and discusses the results obtained from the data collection, point cloud processing, volume estimation, and mass prediction of oil palm fresh fruit bunches (FFB) using RGB-depth data. The main objective was to estimate the mass of oil palm FFBs by multiplying the volume obtained from point cloud data with the average actual density calculated from physical samples. The chapter begins by summarizing the actual measurements of 50 oil palm FFBs, including their mass, volume, and density. It then proceeds to show the process of point cloud volume estimation using CloudCompare software and compares those estimated volumes with actual volumes from water displacement. A correlation analysis between estimated volume and actual mass showed a strong positive relationship, indicating that volume obtained from 2.5D computation is a good predictor of mass. The mass estimation was then performed using the fixed average density, and the predicted results were compared against actual mass values. The validation was further strengthened by using 10 separate samples, where a high Pearson correlation coefficient and R^2 value confirmed the reliability of the proposed method. Each section of this chapter contributes to evaluating the feasibility and effectiveness of using RGB-D data and point cloud processing in estimating oil palm FFB mass.

4.2 Descriptive Statistics Analysis

4.2.1 Actual Mass of Oil Palm FFB

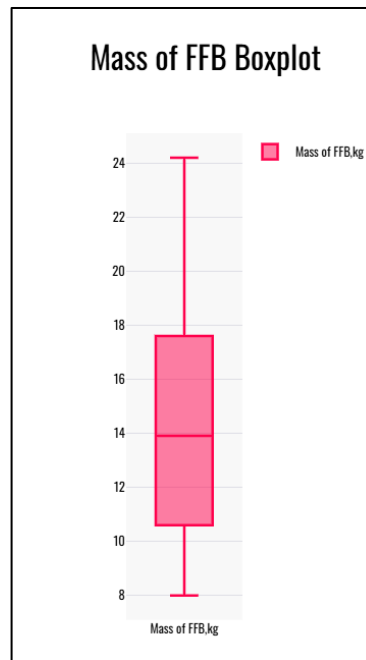


Figure 4.1: Actual mass of FFB (kg) boxplot

Figure 4.1 shows the actual mass of 50 fresh fruit bunch (FFB) samples was measured using a mechanical spring scale and analyzed using descriptive statistics and boxplot visualization. From the data, the masses ranged from 8.0 kg to 24.2 kg, with a mean value of 14.464 kg and a median of 13.9 kg. This indicates that the central tendency of the data lies close to the average, suggesting a relatively balanced distribution of mass across the samples.

The interquartile range (IQR), which represents the middle 50% of the data, was calculated as 7.0 kg, with the first quartile (Q1) at 10.6 kg and the third quartile (Q3) at 17.6 kg. The absence of any detected outliers indicates that all mass values fell

within a statistically acceptable range, suggesting good consistency in the data collection process.

In terms of distribution shape, the skewness value of 0.3835 indicates a slight right skew. The skewness test returned a p-value of 0.255, which means the distribution is potentially symmetrical. The upper boundary ($Q3 + 1.5 \cdot IQR$) was 28.1 kg, and since the maximum observed mass was 24.2 kg, none of the samples exceeded this threshold. This further confirms that no extreme or unusual values were present.

Overall, the distribution of the actual mass of oil palm FFB samples can be described as symmetrical, consistent, and normally distributed, without any extreme variations. This supports the reliability of the physical mass data as a ground truth reference for validating the mass estimation approach in this study.

4.2.2 Actual Size of Oil Palm FFB

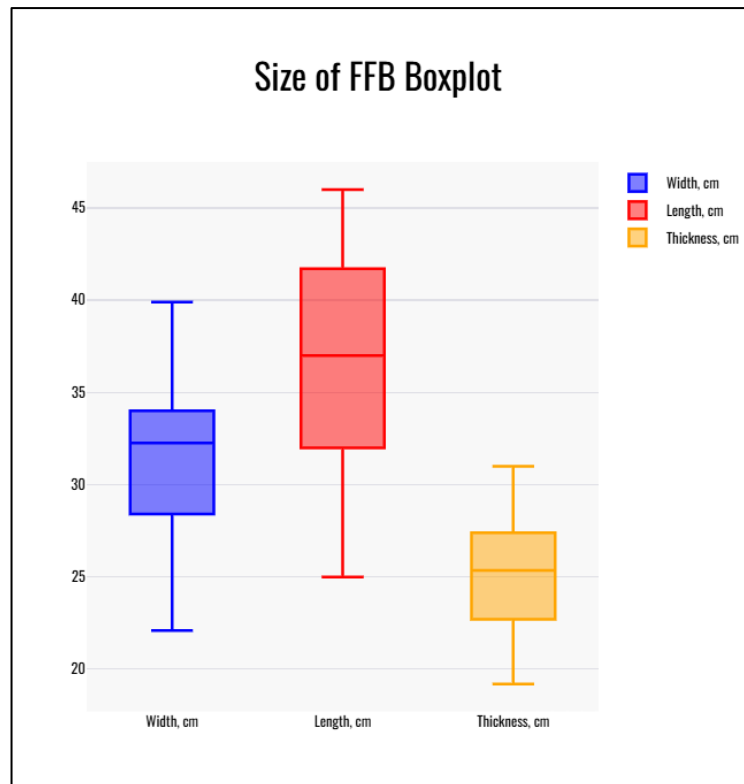


Figure 4.2: Actual Size of oil Palm FFB (cm) boxplot

The physical dimensions of each oil palm fresh fruit bunch (FFB) were measured in three directions which are width, length, and thickness. It is to understand their size distribution and ensure consistency across all samples. Each measurement was statistically analyzed and visualized using boxplots as in **Figure 4.2** to identify trends, spread, and potential irregularities.

The width of the FFB samples ranged from 22.1 cm to 39.9 cm, with a mean of 31.766 cm and a median of 32.25 cm. The interquartile range (IQR) was 5.6 cm, with the first quartile (Q1) at 28.4 cm and the third quartile (Q3) at 34.0 cm. The distribution showed a slight negative skew (skewness = -0.1923), but the p-value of 0.568 indicates that

the data is statistically symmetrical. There were no outliers, and the tails were found to be mesokurtic, suggesting that the spread of the data follows a normal distribution with no extreme variations in the width values.

The length values ranged between 25.0 cm and 46.0 cm, with an average of 36.672 cm and a median of 37.0 cm. The IQR was 9.7 cm, with Q1 at 32.0 cm and Q3 at 41.7 cm. The length distribution was also slightly left-skewed (skewness = -0.1773), but with a p-value of 0.598, it was still considered symmetrical. No outliers were present, and the tails were considered mesokurtic, indicating a normal distribution without heavy extremes.

The thickness of the FFB samples varied from 19.2 cm to 31.0 cm, with a mean of 25.018 cm and a median of 25.35 cm. The IQR was 4.7 cm, with Q1 at 22.7 cm and Q3 at 27.4 cm. The skewness was nearly zero which is -0.03164, with a very high p-value (0.925) confirming a strongly symmetrical distribution. No outliers were detected, suggesting a clean and consistent dataset for this measurement.

Across all three measured dimensions, the FFB samples displayed balanced and symmetrical distributions, with no statistical outliers. This indicates that the physical sizes of the samples were consistently measured and followed a normal distribution pattern. The absence of extreme values also supports the overall quality and representativeness of the collected sample set.

4.2.3 Actual Volume of Oil Palm FFB

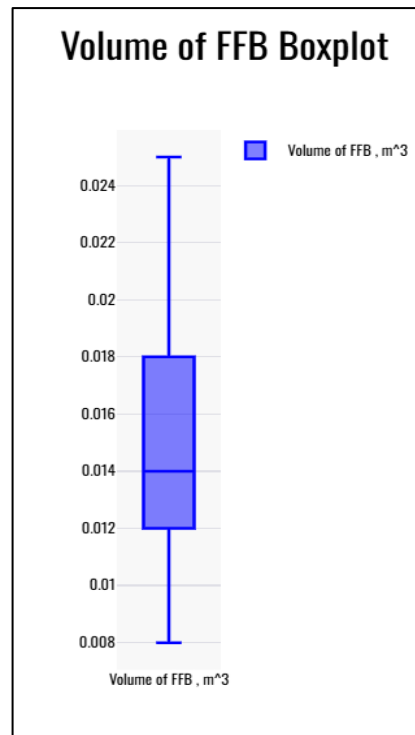


Figure 4.3: Actual Volume of oil palm FFB (m³) boxplot

The actual volume of each oil palm fresh fruit bunch (FFB) was measured using the water displacement method (WDM). The data was analyzed using boxplot statistics as in **Figure 4.3** to evaluate the spread and shape of the volume distribution across all 50 samples.

The measured volume values ranged from 0.008 m³ to 0.025 m³, with a mean of 0.01508 m³ and a median of 0.014 m³. This shows that most of the FFB volumes clustered around the average, and the mean and median being close together suggests a fairly balanced distribution. The interquartile range (IQR) was 0.006 m³, with the first quartile (Q1) at 0.012 m³ and the third quartile (Q3) at 0.018 m³. This indicates

that 50% of the FFB samples had volumes within this narrow range, reflecting consistent size among the majority of the fruits.

The skewness value was 0.3214, indicating a slight right skew in the data. The volume data considered as potentially symmetrical. Importantly, no outliers were detected, meaning all measured volumes fall within the expected statistical boundaries. This shows that the volume measurements are consistent and reliable across the sample set, with no extreme or unusual data points.

In summary, the actual volume of the oil palm FFB samples is normally distributed, symmetrical, and free of outliers, which confirms the accuracy and consistency of the WDM method used. These characteristics also support the use of volume data as a dependable variable in the mass estimation process of this study.

4.2.4 Actual Density of Oil Palm FFB

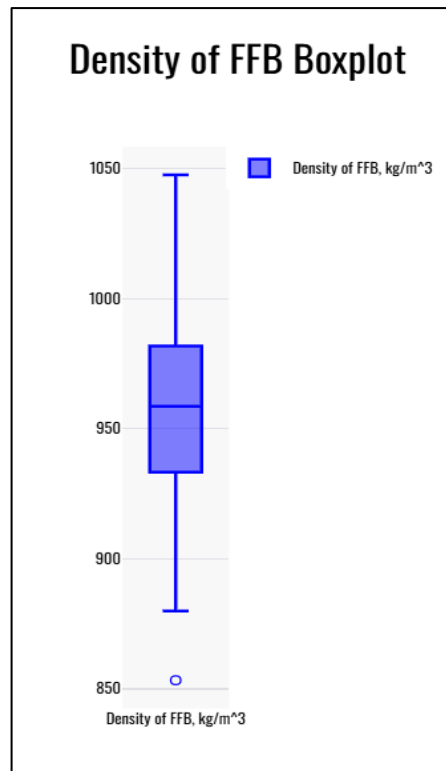


Figure 4.4: Actual Density of Oil Palm FFB (kg/m³) boxplot

Figure 4.4 shows the actual density of each oil palm fresh fruit bunch (FFB) was calculated by dividing its actual mass by its measured actual volume, both obtained through direct methods were analyzed using descriptive statistics and boxplot visualization. This analysis helps to determine the compactness or heaviness of each fruit bunch, which is important factor for estimating mass using volume data.

The observed density values ranged from 853.33 kg/m³ to 1047.62 kg/m³, with a mean of 956.28 kg/m³ and a median of 958.57 kg/m³. The closeness between the mean and median values indicates a fairly balanced distribution. The interquartile range (IQR) was 48.48 kg/m³, showing moderate variability in density among the middle 50% of

samples. The first quartile (Q1) was 933.33 kg/m³, and the third quartile (Q3) was 981.82 kg/m³, meaning most of the FFBs had densities within this range.

The distribution shape was slightly left-skewed, with a skewness value of -0.4086. The data is considered as potentially symmetrical. Only one outlier was detected, which was the lowest value at 853.33 kg/m³. However, this value still falls reasonably close to the lower boundary which is 860.60 kg/m³ and may reflect natural variation in fruit density rather than measurement error.

In conclusion, the actual density values of the oil palm FFB samples show a mostly symmetrical and normal distribution, with only one mild outlier. The consistency in density across the dataset supports its use as a reliable factor in the mass estimation formula. The presence of a normal distribution also ensures that average density is a suitable representation for predicting mass from volume in this study.

4.3 Correlation Analysis Between Actual and Estimated Volume of FFB

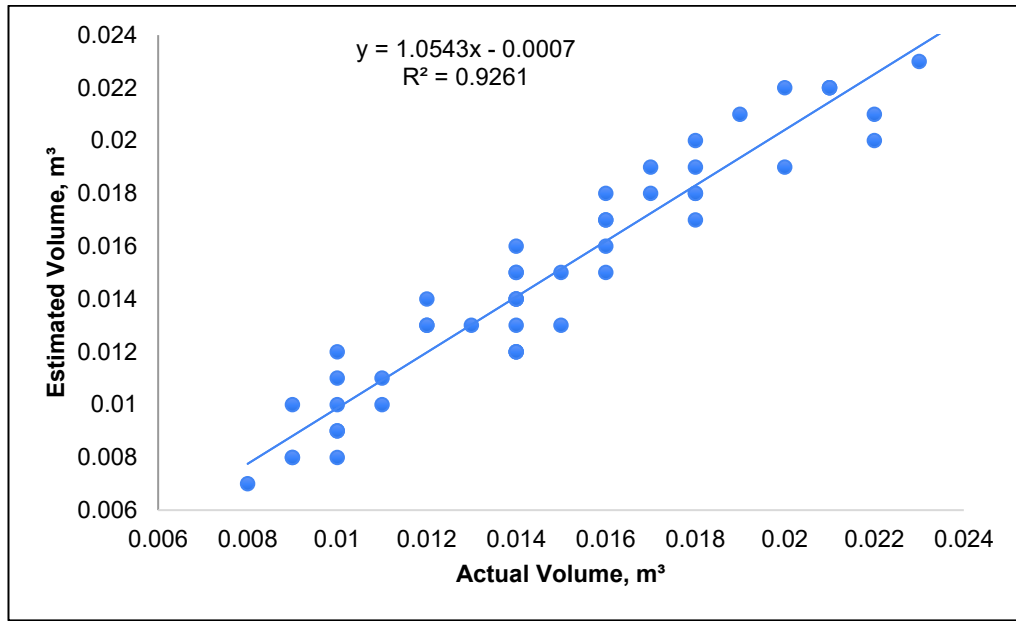


Figure 4.5: Actual Volume of FFB (m³) vs Estimated Volume of FFB (m³)

A scatterplot was developed to visualize the relationship between the actual volume of oil palm fresh fruit bunches (FFB), measured using the water displacement method (WDM), and the estimated volume computed using point cloud processing in CloudCompare. This plot includes all 50 FFB samples and serves to evaluate how closely the volume estimation method matches the actual physical measurements.

From the **Figure 4.5** scatterplot, the data points align closely along a diagonal trend line, suggesting a strong positive linear relationship between the estimated and actual volume values. This was further confirmed by calculating the Pearson correlation coefficient (r), which yielded a value of 0.9623. This value indicates a very strong correlation.

In addition to correlation, the coefficient of determination (r^2) was found to be 0.9261, meaning that approximately 92.61% of the variance in the actual volume data can be explained by the estimated volume generated from the point cloud. This high r^2 value reflects the accuracy and reliability of the estimation process using 2.5D volume computation in CloudCompare. The tight clustering of points along the trend line and the absence of extreme deviations suggest that the estimation method was applied consistently across all samples, with minimal error.

In summary, the scatterplot analysis demonstrates that the estimated volume derived from RGB-depth data is highly consistent with the actual volume measured physically, validating the effectiveness of the point cloud based volume estimation method for oil palm FFB.

4.4 Correlation Analysis Between Actual and Estimated Mass of FFB

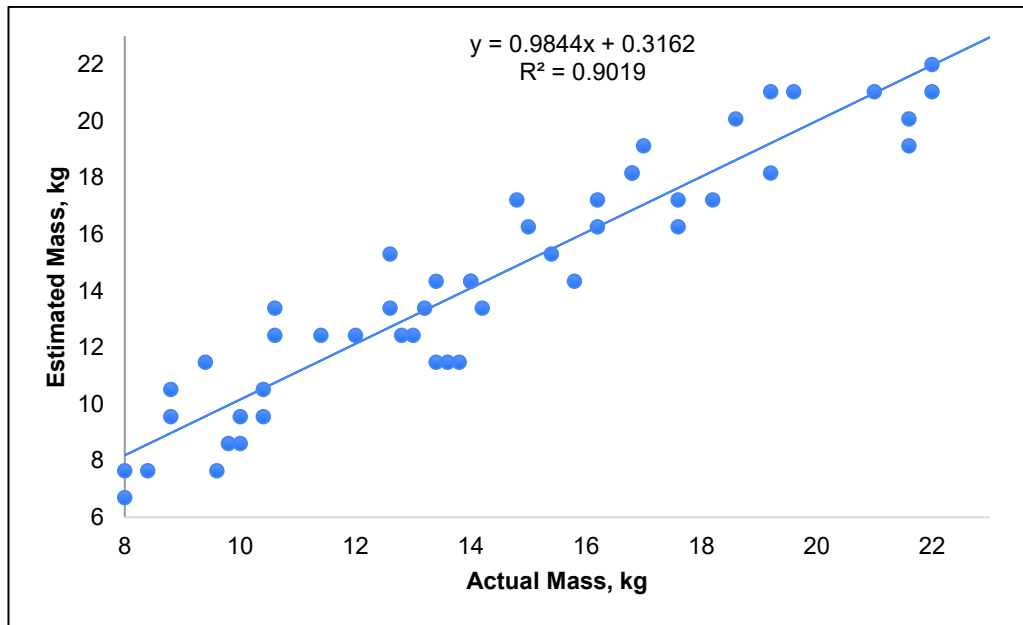


Figure 4.6: Actual Mass of FFB (kg) vs Estimated Mass of FFB (kg)

Figure 4.6 shows a scatterplot was plotted to compare the actual mass of each oil palm fresh fruit bunch (FFB) with its corresponding estimated mass. The actual mass was measured manually using a spring scale, while the estimated mass was calculated using the formula:

$$\text{Estimated Mass} = \rho \times V$$

In this formula, ρ represents the average actual density of all 50 FFB samples, which was determined to be 956.28 kg/m^3 , and V is the estimated volume obtained from CloudCompare using the 2.5D volume computation method. This approach relies on RGB-D point cloud data processed to extract the three-dimensional shape of each FFB.

The scatterplot of actual versus estimated mass revealed that most data points are closely aligned with a linear trend, indicating a strong positive correlation between the two variables. The Pearson correlation coefficient (r) was 0.9496, showing a very strong linear relationship.

Furthermore, the coefficient of determination (r^2) was 0.9019, which means that approximately 90.27% of the variation in actual mass can be explained by the estimated mass values. This high r^2 value reflects the reliability of the estimation method, which is based on accurate volume data and a representative average density.

Overall, the results confirm that the combination of 2.5D volume estimation and a fixed average density provides a reliable technique for predicting FFB mass. This method offers an alternative to manual weighing, which could be beneficial for improving automation and efficiency in palm oil plantations and mills.

4.5 Validation of Mass Estimation using Average Actual Density

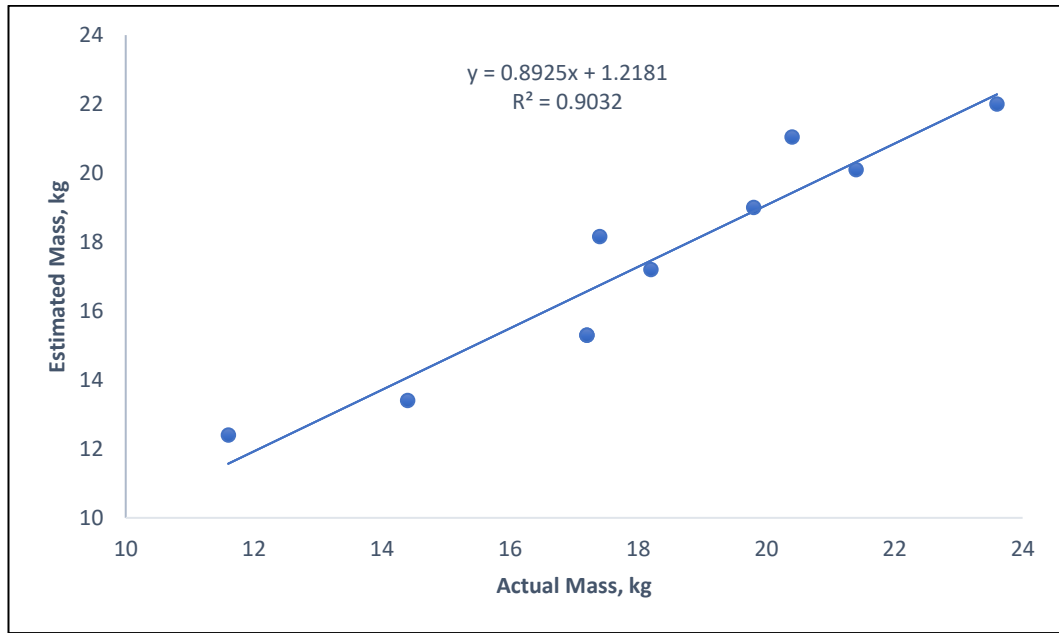


Figure 4.7: Actual Mass of 10 Samples FFB (kg) vs Estimated Mass of 10 Samples FFB (kg)

To evaluate the suitability of using the average actual density from 50 oil palm fresh fruit bunch (FFB) samples for estimating mass, a validation process was carried out using 10 new FFB samples. These validation samples were entirely separate from the initial dataset and had no prior data on mass, volume, or density. The purpose of this validation was to determine whether the fixed average density value of 956.28 kg/m^3 could accurately estimate the mass of oil palm FFBs based solely on volume measurements obtained through point cloud analysis.

The estimated mass for each validation sample was calculated using the formula:

$$\text{Estimated Mass} = \rho \times V$$

Where ρ represents the average actual density of all 50 FFB samples, which was determined to be 956.28 kg/m³, and V is the estimated volume obtained from CloudCompare using the 2.5D volume computation method. These estimated masses were then compared against the actual measured masses of the 10 validation samples to evaluate the performance of the estimation method. **Figure 4.7** shows a scatterplot that was created with actual mass on the x-axis and estimated mass on the y-axis. From this analysis, the Pearson correlation coefficient (r) was found to be 0.9503, and the coefficient of determination (r^2) was calculated as 0.9032.

The Pearson correlation coefficient of 0.9503 indicates a very strong positive linear relationship between actual and estimated mass. Additionally, the r^2 value of 0.9032 suggests that approximately 90.32% of the variation in actual mass can be explained by the estimated mass values based on volume and the average density. These results validate that the use of an average density in the mass estimation process provides a highly reliable approximation when actual mass is unavailable or impractical to measure.

However, some deviations between actual and estimated masses were observed. For instance, a sample had an underestimation of 1.9 kg. These differences may be due to several factors such as natural variation in FFB density caused by differences in ripeness, internal fruit structure, or moisture content. Other possible causes include inaccuracies in volume estimation due to surface noise in point cloud data or mesh gaps during the 2.5D reconstruction process. Additionally, minor errors during manual mass measurement may also have contributed to these differences.

Despite these variations, the high correlation and explanatory power indicated by the r^2 value demonstrate that the average density estimation approach is suitable and practical. It provides a consistent and efficient method for estimating FFB mass without relying on direct physical measurement. This validation supports the feasibility of applying computer vision and point cloud techniques in oil palm FFB mass estimation.

4.6 Summary

In summary, the findings from Chapter 4 support the practicality of using RGB-depth data combined with point cloud volume estimation to predict the mass of oil palm FFBs. The actual mass, volume, and density data from 50 samples were successfully recorded and used to calculate an average density. This average was then applied to volume estimations derived from 2.5D meshing in CloudCompare to estimate mass. The comparison between estimated and actual values demonstrated a strong correlation, suggesting that the method is suitable for non-destructive and contactless mass prediction. The validation using 10 new samples further confirmed the accuracy of this approach, with a Pearson correlation coefficient of 0.9504 and an R^2 of 0.903. Although some variation existed between predicted and actual masses, the differences were within acceptable limits and could be explained by natural variation in fruit properties. Overall, the results indicate that using a fixed average density in combination with volume estimated from RGB-D point cloud data is an effective approach for oil palm FFB mass estimation in practical applications.

CHAPTER 5

CONCLUSION

5.1 Conclusions

This study was conducted to investigate the feasibility of using RGB-Depth (RGB-D) data for estimating the volume of oil palm Fresh Fruit Bunches (FFB) as a basis for mass estimation. The first objective, which was to determine the density of oil palm FFB, was achieved through direct physical measurements using a mechanical scale and water displacement method. The second objective, which focused on identifying the variation between the estimated and measured volume of FFB, was fulfilled by processing RGB-D videos into point cloud models and estimating volume using CloudCompare software.

In addition to achieving both objectives, this study also examined the relationship between the RGB-D estimated volume and the actual mass of the FFB. A strong positive correlation was found, indicating that volume data obtained from RGB-D imaging can be reliably used to estimate mass when paired with average density values. This confirms that RGB-D data is not only feasible but also effective as a non-contact method for early mass estimation in oil palm operations.

Overall, this project successfully showed that RGB-D imaging has the potential to support practical applications in oil palm plantations by enabling faster and more consistent estimation of FFB mass. This method may help reduce processing delays,

improve harvest planning, and support better yield estimation, especially in large-scale plantation environments.

5.2 Recommendations

This study has provided valuable insights into the use of RGB-Depth (RGB-D) data for estimating the mass of oil palm Fresh Fruit Bunches (FFB). In line with the findings and limitations observed, the following recommendations are proposed to improve future research and potential applications.

Firstly, future work should explore the integration of machine learning or deep learning models to develop a more automated and intelligent mass prediction system. By using a larger dataset and training models on the relationship between RGB-D volume and actual mass, it may be possible to improve accuracy and enable real-time predictions in the field.

Secondly, it is recommended to test the system under different environmental conditions, such as varying lighting, weather, and background noise. This will help ensure the method is robust and suitable for real-world deployment across various plantation settings, especially for outdoor applications.

Lastly, future studies should consider increasing the sample size and including oil palm FFB from different plantation blocks or age groups. This will help evaluate how fruit variability affects the accuracy of the volume and mass estimation system and allow for better generalization across different harvesting conditions.

5.3 Assessment on Implication on Society and Environment or Impacts on Society, Health, and Safety

This project, which applies RGB-Depth (RGB-D) data to estimate the mass of oil palm Fresh Fruit Bunches (FFB), has several implications on society, safety, and the environment. In current plantation practice, manual weighing requires physical lifting of heavy fruit bunches, which can lead to worker fatigue and injury. The proposed non-contact method reduces physical strain, contributing to a safer and more efficient work environment.

From an environmental perspective, accurate and timely mass estimation supports better harvest planning, reducing delays between harvesting and milling. This helps maintain fruit quality, improves oil extraction, and minimizes waste, contributing to more sustainable plantation practices.

On a societal level, the adoption of sensor-based technology encourages the shift towards smart agriculture. It reduces dependency on manual labour, increases operational efficiency, and opens up opportunities for technical training and job creation in agricultural technology. These developments support both large-scale plantations and smallholder farmers.

In summary, this system promotes safer working conditions, improved harvest efficiency, and better environmental management, aligning with the goals of sustainable and technology-driven agriculture.

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APPENDICES

Appendix A: Actual Dataset of FFB

Table A: Actual Dataset of Oil Palm FFB

No.	Mass of FFB, kg	Size, cm			Initial Water Level, l	Final Water Level, l	Volume of FFB, l	Density of FFB, kg/l
		Width	Length	Thickness				
1	19.2	37	46	30	70	91	21	0.914285714
2	12.8	33	30.4	25	70	85	15	0.853333333
3	17.6	39.9	44	28.5	70	88	18	0.977777778
4	17.6	33.7	45.4	26.2	70	88	18	0.977777778
5	18.6	35.5	43.5	28.7	70	89	19	0.978947368
6	24.2	32	45.5	28	70	95	25	0.968
7	10.4	26	37	23	70	81	11	0.945454545
8	14	30	37.6	21.5	70	85	15	0.933333333
9	8	25	36	31	70	78	8	1
10	17	35	44	24	70	88	18	0.944444444
11	14	32	41.2	24	70	84	14	1
12	21.6	34	39.4	27.8	70	92	22	0.981818182

13	16.8	31.2	38.7	27.4	70	88	18	0.933333333
14	21	38.4	41.7	28.3	70	91	21	1
15	15	30.4	39.4	25.5	70	86	16	0.9375
16	19.2	35.7	37	25.3	70	90	20	0.96
17	13.4	33.6	29.8	27.4	70	84	14	0.957142857
18	9.6	32.5	29.6	22.2	70	80	10	0.96
19	19.6	33.7	40.4	26.1	70	90	20	0.98
20	21.6	31.6	41.5	27.4	70	92	22	0.981818182
21	9.4	30	34.6	22.7	70	80	10	0.94
22	22	39.1	43.6	28.7	70	93	23	0.956521739
23	15.8	36.7	42	23.7	70	86	16	0.9875
24	16.2	37.8	43	26.8	70	86	16	1.0125
25	18.2	37.4	42.1	29	70	88	18	1.011111111
26	13.4	33.8	40.1	21.4	70	84	14	0.957142857
27	16.8	33.8	40.8	25.7	70	87	17	0.988235294
28	10	34	29.5	23.4	70	80	10	1
29	14.8	32.4	39.1	25.4	70	86	16	0.925
30	22	39.5	44	28.1	70	91	21	1.047619048
31	13.8	33.8	32.5	26	70	84	14	0.985714286
32	9.8	27	29.5	19.2	70	80	10	0.98
33	12	25.7	25	22.3	70	83	13	0.923076923
34	12.6	26.8	27.6	25.4	70	84	14	0.9

35	13	30.7	34.3	23.7	70	84	14	0.928571429
36	8	26.1	31.1	20.6	70	79	9	0.888888889
37	10.6	28.1	33.7	22.2	70	82	12	0.883333333
38	12.6	35.1	38.5	26.4	70	84	14	0.9
39	8.4	26.6	36.2	24.2	70	79	9	0.933333333
40	10.6	31.5	32	21.6	70	82	12	0.883333333
41	11.4	33.1	36.1	27	70	82	12	0.95
42	8.8	28.4	33	21.3	70	79	9	0.977777778
43	10.4	33.4	30.1	26.7	70	81	11	0.945454545
44	15.4	28.4	43.2	24.5	70	86	16	0.9625
45	13.6	32.1	35.5	23.6	70	84	14	0.971428571
46	10	23.3	28	21.7	70	80	10	1
47	16.2	26.5	32.6	21.6	70	87	17	0.952941176
48	13.2	29.8	28.4	27.2	74	88	14	0.942857143
49	14.2	25.1	32.6	23.4	74	88	14	1.014285714
50	8.8	22.1	26.8	20.1	74	84	10	0.88

Appendix B: Actual Volume vs Estimated Volume Dataset of FFB

Table B: Actual Volume vs Estimated Volume Dataset of FFB

No. of FFB	Actual Volume of FFB, m ³	Estimated Volume from CloudCompare, m ³
1	0.021	0.022
2	0.015	0.013
3	0.018	0.017
4	0.018	0.018
5	0.019	0.021
6	0.025	0.026
7	0.011	0.01
8	0.015	0.015
9	0.008	0.007
10	0.018	0.02
11	0.014	0.015
12	0.022	0.02
13	0.018	0.019
14	0.021	0.022

15	0.016	0.017
16	0.02	0.019
17	0.014	0.015
18	0.01	0.008
19	0.02	0.022
20	0.022	0.021
21	0.01	0.012
22	0.023	0.023
23	0.016	0.015
24	0.016	0.017
25	0.018	0.018
26	0.014	0.012
27	0.017	0.019
28	0.01	0.01
29	0.016	0.018
30	0.021	0.022
31	0.014	0.012
32	0.01	0.009

33	0.013	0.013
34	0.014	0.016
35	0.014	0.013
36	0.009	0.008
37	0.012	0.013
38	0.014	0.014
39	0.009	0.008
40	0.012	0.014
41	0.012	0.013
42	0.009	0.01
43	0.011	0.011
44	0.016	0.016
45	0.014	0.012
46	0.01	0.009
47	0.017	0.018
48	0.014	0.014
49	0.014	0.014
50	0.01	0.011

Appendix C: Actual Mass vs Estimated Mass of FFB Dataset

Table C: Actual Mass vs Estimated Mass of FFB Dataset

No. of FFB	Estimated Volume from CloudCompare, m ³	Average Density of 50 samples FFB, kg/m ³	Estimated Mass, kg	Actual Mass of FFB
1	0.022	956.28	21.04	19.2
2	0.013	956.28	12.43	12.8
3	0.017	956.28	16.26	17.6
4	0.018	956.28	17.21	17.6
5	0.021	956.28	20.08	18.6
6	0.026	956.28	24.86	24.2
7	0.01	956.28	9.56	10.4
8	0.015	956.28	14.34	14
9	0.007	956.28	6.69	8

10	0.02	956.28	19.13	17
11	0.015	956.28	14.34	14
12	0.02	956.28	19.13	21.6
13	0.019	956.28	18.17	16.8
14	0.022	956.28	21.04	21
15	0.017	956.28	16.26	15
16	0.019	956.28	18.17	19.2
17	0.015	956.28	14.34	13.4
18	0.008	956.28	7.65	9.6
19	0.022	956.28	21.04	19.6
20	0.021	956.28	20.08	21.6
21	0.012	956.28	11.48	9.4
22	0.023	956.28	22	22
23	0.015	956.28	14.34	15.8
24	0.017	956.28	16.26	16.2
25	0.018	956.28	17.21	18.2
26	0.012	956.28	11.48	13.4

27	0.019	956.28	18.17	16.8
28	0.01	956.28	9.56	10
29	0.018	956.28	17.21	14.8
30	0.022	956.28	21.04	22
31	0.012	956.28	11.48	13.8
32	0.009	956.28	8.61	9.8
33	0.013	956.28	12.43	12
34	0.016	956.28	15.3	12.6
35	0.013	956.28	12.43	13
36	0.008	956.28	7.65	8
37	0.013	956.28	12.43	10.6
38	0.014	956.28	13.39	12.6
39	0.008	956.28	7.65	8.4
40	0.014	956.28	13.39	10.6
41	0.013	956.28	12.43	11.4
42	0.01	956.28	9.56	8.8
43	0.011	956.28	10.52	10.4

44	0.016	956.28	15.3	15.4
45	0.012	956.28	11.48	13.6
46	0.009	956.28	8.61	10
47	0.018	956.28	17.21	16.2
48	0.014	956.28	13.39	13.2
49	0.014	956.28	13.39	14.2
50	0.011	956.28	10.52	8.8

Appendix D: Validation of 10 Samples of FFB

Table D: Validation of 10 Samples of FFB Dataset

No. of FFB	Estimated Volume from CloudCompare, m ³	Average Density, kg/m ³	Estimated Mass of FFB, kg	Actual Mass of FFB, kg	Difference,kg
1	0.013	956.28	12.4	11.6	0.8
2	0.018	956.28	17.2	18.2	-1
3	0.016	956.28	15.3	17.2	-1.9
4	0.019	956.28	18.16	17.4	0.76
5	0.022	956.28	21.04	20.4	0.64
6	0.022	956.28	22	23.6	-1.6
7	0.014	956.28	13.4	14.4	-1
8	0.016	956.28	15.3	17.2	-1.9
9	0.02	956.28	19	19.8	-0.8
10	0.021	956.28	20.1	21.4	-1.3

Appendix E: Figure of All Oil Palm Fresh Fruit Bunch (FFB)



Figure E.1: FFB No.1; Mass: 19.2 kg,
Volume:21 l



Figure E.2: FFB No.2; Mass: 12.8 kg, Volume:
15 l



Figure E.3: FFB No.3; Mass: 17.6 kg ,
Volume: 18 l



Figure E.4: FFB No.4; Mass: 17.6 kg ,
Volume: 18 l



Figure E.5: FFB No.5; Mass: 18.6 kg ,
Volume: 19 l



Figure E.6: FFB No.6; Mass: 24.2 kg ,
Volume: 25 l

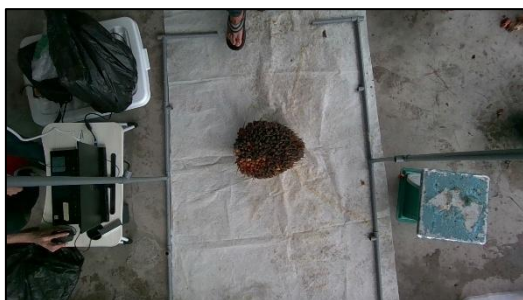


Figure E.7: FFB No.7; Mass: 10.4 kg ,
Volume: 11 l

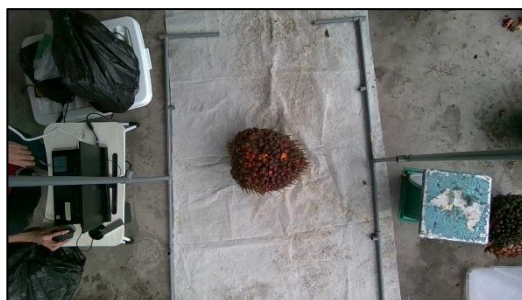


Figure E.8: FFB No.8; Mass: 14 kg ,
Volume: 15 l

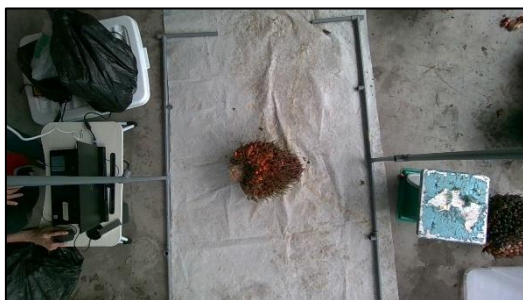


Figure E.9: FFB No.9; Mass: 8 kg ,
Volume: 8 l

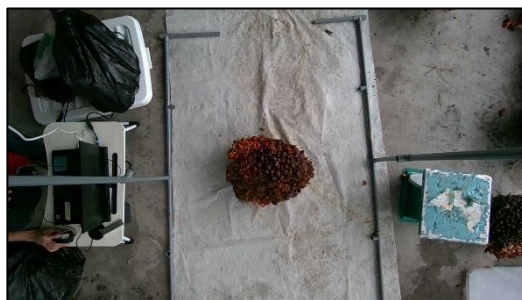


Figure E.10: FFB No.10; Mass: 17 kg ,
Volume: 18 l



Figure E.11: FFB No.11; Mass: 14 kg ,
Volume: 14 l

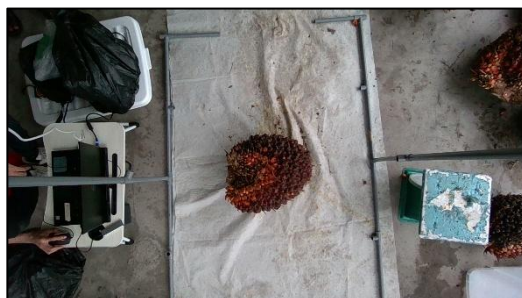


Figure E.12: FFB No.12; Mass: 21.6 kg,
Volume: 22 l

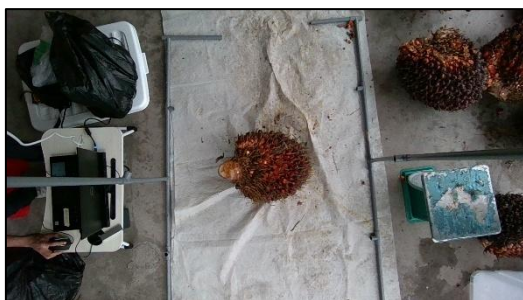


Figure E.13: FFB No.13; Mass: 16.8 kg ,
Volume: 18 l



Figure E.14: FFB No.14; Mass: 21 kg ,
Volume: 21 l

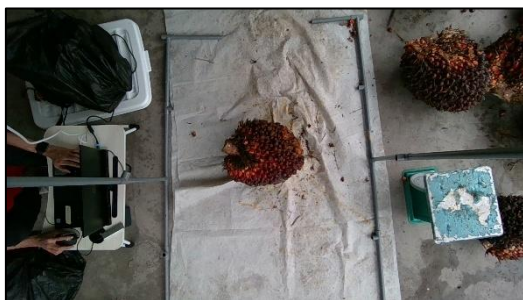


Figure E.15: FFB No.15; Mass: 15 kg ,
Volume: 16 l



Figure E.16: FFB No.16; Mass: 19.2 kg,
Volume: 20 l

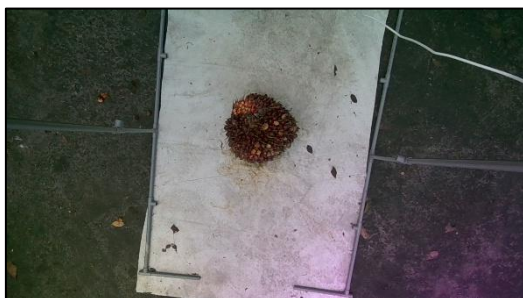


Figure E.17: FFB No.17; Mass: 13.4 kg ,
Volume: 14 l

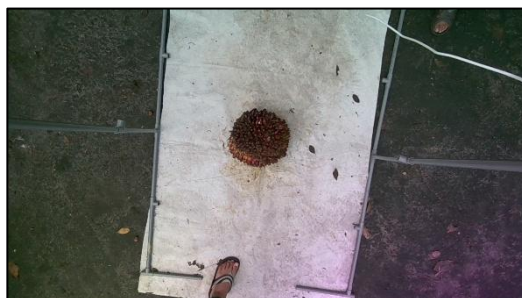


Figure E.18: FFB No.18; Mass: 9.6 kg ,
Volume: 10 l



Figure E.19: FFB No.19; Mass: 19.6 kg ,
Volume: 20 l



Figure E.20: FFB No.20; Mass: 21.6 kg ,
Volume: 22 l



Figure E.21: FFB No.21; Mass: 9.4 kg ,
Volume: 10 l



Figure E.22: FFB No.22; Mass: 22 kg ,
Volume: 23 l



Figure E.23: FFB No.23; Mass: 15.8 kg ,
Volume: 16 l



Figure E.24: FFB No.24; Mass: 16.2 kg ,
Volume: 16 l



Figure E.25: FFB No.25; Mass: 18.2 kg ,
Volume: 18 l



Figure E.26: FFB No.26; Mass: 13.4 kg ,
Volume: 14 l

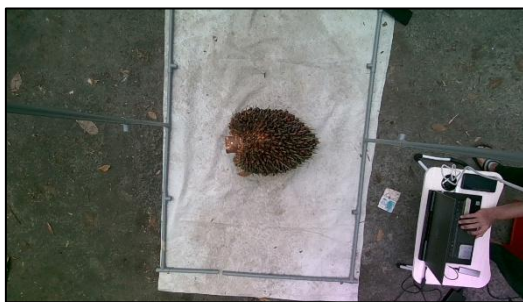


Figure E.27: FFB No.27; Mass: 16.8 kg ,
Volume: 17 l

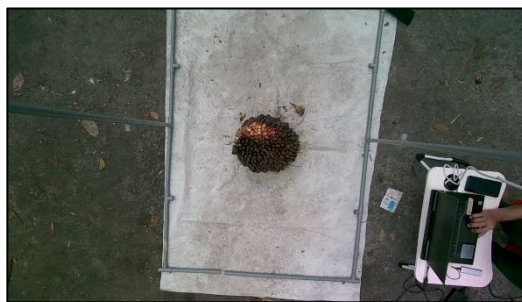


Figure E.28: FFB No.28; Mass: 10 kg ,
Volume: 10 l

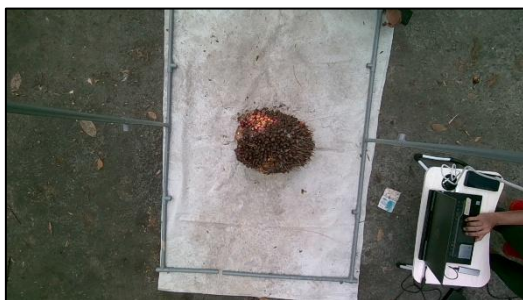


Figure E.29: FFB No.29; Mass: 14.8 kg ,
Volume: 16 l

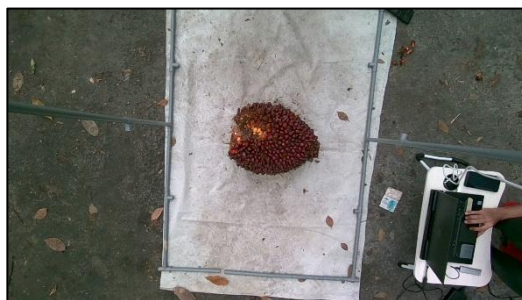


Figure E.30: FFB No.31; Mass: 22 kg ,
Volume: 21 l



Figure E.31: FFB No.31; Mass: 13.8 kg ,
Volume: 14 l



Figure E.32: FFB No.32; Mass: 9.8 kg ,
Volume: 10 l



Figure E.33: FFB No.33; Mass: 12 kg ,
Volume: 13 l



Figure E.34: FFB No.34; Mass: 12.6 kg ,
Volume: 14 l



Figure E.35: FFB No.35; Mass: 13 kg ,
Volume: 14 l



Figure E.36: FFB No.36; Mass: 8 kg ,
Volume: 9 l



Figure E.37: FFB No.37; Mass: 10.6 kg ,
Volume: 12 l



Figure E.38: FFB No.38; Mass: 12.6 kg ,
Volume: 14 l



Figure E.39: FFB No.39; Mass: 8.4 kg ,
Volume: 9 l



Figure E.40: FFB No.40; Mass: 10.6 kg ,
Volume: 12 l



Figure E.41: FFB No.41; Mass: 11.4 kg ,
Volume: 12 l



Figure E.42: FFB No.42; Mass: 8.8 kg ,
Volume: 9 l



Figure E.43: FFB No.43; Mass: 10.4 kg ,
Volume: 11 l



Figure E.44: FFB No.44; Mass: 15.4 kg ,
Volume: 16 l



Figure E.45: FFB No.45; Mass: 13.6 kg ,
Volume: 14 l



Figure E.46: FFB No.46; Mass: 10 kg ,
Volume: 10 l



Figure E.47: FFB No.47; Mass: 16.2 kg ,
Volume: 17 l



Figure E.48: FFB No.48; Mass: 13.2 kg ,
Volume: 14 l



Figure E.49: FFB No.49; Mass: 14.2 kg ,
Volume: 14 l



Figure E.50: FFB No.50; Mass: 8.8 kg ,
Volume: 10 l